

Alternative Asymptotics and Fixed-Alternative Power Analysis ^{*}

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Abstract

When constructing Wald tests, consistency is the key property required for the variance estimator. This property ensures asymptotic validity of Wald tests and confidence intervals. Classical efficiency comparisons of hypothesis tests indicate all consistent variance estimators lead to equivalent Wald tests under local power approximations. This paper develops a simple asymptotic framework under fixed alternatives, which leads to new conclusions. In particular, we identify that variance estimation will have a first-order impact on the efficiency of Wald tests when size tends to zero with sample size while effect sizes are fixed. We apply this framework to several applications, including cluster-robust inference and quantile regression. In the case of cluster-robust inference, we provide for an asymptotic framework in which choosing the wrong level of clustering can lead different power properties asymptotically. Numerical evidence suggests the results are applicable to moderate sample sizes and that the results can provide a useful approximation in standard ranges of size and power.

1 Introduction

Much of empirical work in economics follows a three step recipe: estimate the parameter of interest, estimate the asymptotic variance, then construct a test statistic or confidence interval to answer the research question. The first step is generally treated differently than the other two; while discussions on parameter estimation often focus on efficiency, the dialogue around variance estimation and testing typically focuses on robustness to misspecification. Ignoring any efficiency implications of variance estimation is at odds with the lived experience of empirical researchers, in that robust variance estimators are often considered “conservative.” In this paper we provide a first-order asymptotic theory which characterises scenarios in which variance estimation affects the asymptotic power of tests. The resulting asymptotic theory provides a theoretical foundation for several common “folk” theorems in applied work. This paper also clarifies the setting in which variance estimation is important for efficiency considerations: when the alternative is large and size is small.

The choice of variance estimator is an ever present decision in applied work. There is a menu of available robust, consistent variance estimators in standard statistical packages. Researchers with grouped data must determine whether to compute cluster-robust standard errors, and what level to cluster at. In the case of quantile regression, researchers choose a kernel density estimator to use. In likelihood settings under correct

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specification, the Fisher information matrix can be estimated using the outer product of the score or the second derivative of the log-likelihood. Robust variance estimators in time series involve choosing a kernel and truncation point. We will not consider all these examples here, however we provide a framework that is suited to studying the effect of variance estimation in many of these contexts.

This paper makes three key contributions to the econometric literature on hypothesis testing. First, we clarify the settings in which variance estimation can impact the power of tests. When the alternative is large relative to sample size and size tends to 0, test statistics using different variance estimators will have different first-order asymptotic properties. The limiting power along the sequence of tests depends not only on consistency of the variance estimator, but also the variance and possible covariance with the estimator of the parameter of interest.

Second, we present some new insights on sample size calculation and connections to Bahadur asymptotic relative efficiency. In the Bahadur setting, size tends to zero as sample size increases, and power converges to a constant. We point out that this setting in fact does not lend itself well to sample size calculations as the limiting power drops out of the analysis. Our numerical evidence suggests that our approach is competitive with local power when we aim to find the minimum sample size to achieve a certain power under a particular alternative.

Last, we apply this approach to several applications, including smooth generalized method of moments (GMM) problems, linear models with possible cluster dependence, and quantile regression.

The theory developed in this paper takes a different approach compared with the traditional local-asymptotic theory of Pitman (1949), Engle (1984), Newey and McFadden (1994), van der Vaart (1998), and Le Cam (2012). That work finds that a broad class of tests statistics have the same limiting distribution under local-alternatives. Our analysis is non-local, which leads to these equivalencies no longer holding in general. Under fixed alternatives, we can view a re-scaled version of the tests statistic as an estimator of a non-centrality parameter: the ratio of the effect size to the variance. With this perspective in mind, the asymptotic behaviour of the test statistic depends on both the estimation of the numerator (depending on the parameter estimator) and the denominator (depending on the variance estimator). In the case of Wald tests, local equivalence holds whenever the same parameter estimator is used in two different tests, even if different consistent variance estimators are used. This equivalence no longer holds in our asymptotic theory when different estimators of the asymptotic variance are used. It turns out that variance estimation has an impact on asymptotic power in testing regimes where size converges to 0 and power converges to some fixed constant. These asymptotics have a similar flavour to the asymptotic experiment proposed in Bahadur (1967). A benefit of our approach is that no large-deviation results are necessary. In a similar fashion, our approach is also more broadly applicable than the measure proposed in Hodges and Lehmann (1956), where size converges to a constant and power converges to 1, an approach which also requires large-deviation theory.

We are not the first to use asymptotics to assess the behaviour of different variance estimators. Sun, Phillips, and Jin (2008) compares fixed-bandwidth to small bandwidth asymptotics in constructing standard errors for time-series regressions. They note that traditional optimal choices of the bandwidth are chosen based on estimation criteria, rather than testing criteria, and develop a theory where the power and size of tests depend explicitly on bandwidth choice in a variety of asymptotic frameworks. Iacone, Leybourne, and Taylor (2013) uses fixed-b asymptotics to choose between tuning parameters for variance estimators and test statistics when testing for breaks in a linear trend. This paper links the estimation of the variance to the power of tests along a different sequence of tests where size tends to zero. In Kato (2012), the asymptotic

distribution is derived for the kernel density estimator for the asymptotic variance in the quantile regression setting of Koenker and Bassett (1978), for the particular choice of a uniform kernel. We extend the results in Kato (2012), and link the form of the asymptotic bias and variance to power calculations.

There has also been other work in econometrics on ways of characterising the behaviour of tests under the alternative that also differ from the local asymptotic normality approach of Le Cam (2012). Kim and Perron (2009) propose using an approximate version of the Bahadur (1967) asymptotic relative efficiency measure when comparing tests for structural breaks in time series. Canay and Otsu (2012) used Hodges-Lehmann asymptotic relative efficiency to assess the efficiency of GMM and generalized empirical likelihood tests of moments conditions. A benefit of our approach is broad applicability to testing problems most frequently encountered in empirical work, while maintaining an exact asymptotic comparison. The approach in this paper is perhaps most similar to Bahadur (1960a), Bahadur (1960b), and Bahadur (1967). In that setting, the rate at which p-values tend to zero under an alternative are connected to the rate at which size can be sent to zero and a fixed non-trivial power can be attained asymptotically. As discussed in Groeneboom and Oosterhoff (1981), this approach presents some challenge for a practical approach to asymptotical relative efficiency in that the limiting non-trivial power does not show up in the analysis of Bahadur asymptotic relative efficiency. We do not make claims on ranking tests using our power analysis, as there is no guarantee in the general case that size tends to zero at the same rate across tests. On the other hand, our analysis provides for non trivial power calculations when size is small and the alternative is large relative to sample size.

We demonstrate the broad applicability of our approach by considering several important applications of the theory. We derive an asymptotic approximation of Wald test statistics for general use in GMM problems. The main result is quite general, allowing for cluster-dependence and non-smooth moment conditions. The first specific application we provide is to cluster-robust inference in the linear model. The general framework for cluster-robust inference we adopt is that in Hansen and Lee (2019). An alternative approach, the design-based approach, is discussed in Abadie, Athey, Imbens, and Wooldridge (2020). Popularized in Bertrand, Duflo, and Mullainathan (2004), some recent work in econometrics has focused on the choice of cluster level. In Cameron and Miller (2015) it is argued that the coarsest cluster level should always be used. Abadie, Athey, Imbens, and Wooldridge (2023) presents a design-based approach to choosing the appropriate cluster level, along with some finite-sample results. MacKinnon, Nielsen, and Webb (2023) provide a sequential testing procedure to detect the correct clustering level, within the model-based framework we adopt. We show that there is an unambiguous loss of efficiency when independent observations are included in the same cluster. Our results imply a method for researchers to conduct power analysis to see if the efficiency loss in their case is severe, or if there is little to be lost from the added robustness.

Our second specific application is to quantile regression. We focus on the linear conditional quantile regression model of Koenker and Bassett (1978). In this case, classic approaches to variance estimation involve estimators of the conditional density of the error term. We focus on the kernel density estimator of Powell (1991). The default choices in the `quantreg` package in R and the `qreg` function in Stata are the Gaussian and Epanechnikov kernel, respectively. We show that the choice of kernel and bandwidth do have an impact on the first-order behaviour of the test statistics in our setting.

Our distributional theory extends results in Bentkus, Jing, Shao, and Zhou (2007), Omey and van Gulck (2009), and Shao and Zhang (2009), where one-sample t-statistics and similar types of statistics are considered. We extend the basic theory to GMM problems under sampling with cluster-dependence, including non-smooth problems such as quantile regression. In Bentkus et al. (2007) these asymptotic distributions are

used to motivate asymptotic power functions. We use this type of calculation to motivate our own relative efficiency comparison.

The rest of the paper proceeds as follows: we start by introducing the principles of our analysis in Section 2, in the context of a simple testing problem: hypothesis testing for means. In Section 3, we provide a treatment of the distribution of Wald statistics in GMM settings, under fixed alternatives. A discussion on using the framework for power analysis is discussed in Section 4 and some simulations are provided in Section 5 to show the efficacy of the methods here in making finite-sample predictions. A summary of our results is discussed in Section 6. Throughout, Φ is used to denote the standard normal cumulative distribution function and $\Rightarrow$ is used to denote convergence in distribution. $C_{1-\alpha}^2$ and C_β are used to denote the $1 - \alpha$ quantile of a χ_1^2 random variable and the β quantile of a standard normal random variable respectively. We use the standard stochastic order symbols O_P and o_P to mean bounded in probability and converges to zero in probability respectively.

2 An Illustrative Example: the Sample Mean

We begin by considering a simple testing problem: a two sided hypothesis test for the sample mean. To illustrate the basic approach, we compare the classic Wald test statistic with a cluster-robust version. Under sequences of local-alternatives, these test statistics have the same asymptotic properties, and therefore the same asymptotic power function. Hence, it is not possible to use asymptotic methods to compare such tests. To overcome this, we compare the asymptotic distributions under fixed alternatives. In this asymptotic setting we find that the asymptotic distributions differ for the two test statistics. This leads to a natural relative efficiency comparison, in which we find that when the observations are independent (i.e. both test statistics have correct asymptotic size) there is an asymptotic power loss associated with using the cluster-robust test statistic.

2.1 Cluster-robust inference

Consider a sample $\{X_{gi}\}$, where i denotes observation i in group g . There are G equal-sized groups, each containing n_g observations, for a total of $Gn_g = n$ observations. For the purposes of this example, we consider the case that n_g is fixed, so that $n/G = n_g$ for all sample sizes.¹ A concerned researcher suggests that we should use cluster-robust methods since the data were grouped when collected, however we know that the observations are independent and identically distributed. For all g, i , we have that $\mathbb{E} X_{gi} = \mu$ and $\text{Var}(X_{gi}) = \sigma^2$. We would like to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. We construct Wald tests based on the sample mean:

$$\bar{X}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} X_{gi}.$$

We compare this test statistic, the classic Wald test-statistic, to a cluster-robust version.² The classic Wald test statistic is

$$W_h = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_h^2}, \quad \hat{\sigma}_h^2 = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} (X_{gi} - \bar{X}_n)^2. \quad (2.1)$$

¹We can also accommodate unbalanced designs with growing cluster sizes; this type of result is also covered in Section 3.

²For simplicity, we do not include any degrees-of-freedom correction, whether for the variance estimator or number of clusters. Since we use the large- G asymptotics of Hansen and Lee (2019), these degrees of freedom corrections disappear in the limit.

The cluster-robust Wald statistic is

$$W_c = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_c^2}, \quad \hat{\sigma}_c^2 = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{gi} - \bar{X}_n) \right)^2. \quad (2.2)$$

where since we will have $G \rightarrow \infty$ the estimator is still consistent for σ^2 .

Under the null hypothesis, and without any cluster dependence, we have that each statistic has the same asymptotic distribution,

$$\begin{aligned} W_h &\Rightarrow \chi_1^2 \\ W_c &\Rightarrow \chi_1^2 \end{aligned}$$

which follows from a basic application of Slutsky's theorem. The numerator of each test statistic, divided by σ^2 , is asymptotically χ_1^2 . In the denominator of each test statistic is a consistent estimator of the variance. This first-order analysis effectively considers the properties of the test statistics under the simplifying assumption that the variance estimator is equal to its probability limit.

The same logic holds in the case of a sequence of local alternatives. Consider $\mu_n = \mu_0 + \delta/\sqrt{n}$. In this case, Slutsky's theorem applies again: the only change is that the numerator of the test statistic is no longer correctly centered, therefore the limiting distribution for each test-statistic is $\chi_1^2(\delta^2/\sigma^2)$. Local power analysis cannot distinguish between a test statistic constructed using $\hat{\sigma}_h^2$ and one constructed using $\hat{\sigma}_c^2$. This asymptotic equivalence leads to the common conclusion among econometricians that the choice of variance estimator does affect the power of the test.

Now consider the fixed alternative $\mu = \Delta + \mu_0$. We are testing $H_0 : \mathbb{E} X_{ig} = \mu_0$. Define the non-centrality parameter:

$$\xi := \frac{\Delta}{\sigma}.$$

For $a \in \{h, c\}$, expansions of the test statistics under a fixed alternative gives

$$\frac{1}{n} W_a = \frac{(\bar{X}_n - \mu)^2}{\hat{\sigma}_a^2} + \frac{2\Delta(\bar{X}_n - \mu)}{\hat{\sigma}_a^2} + \frac{\Delta^2}{\hat{\sigma}_a^2}. \quad (2.3)$$

The first two terms converge in probability to 0, and the last term converges to ξ^2 in each case, and thus we can say that $\frac{1}{n} W_a \xrightarrow{P} \xi^2$. Thus, one way of viewing the test statistic under a fixed alternative is as a scaled estimator of the non-centrality parameter ξ^2 . In (2.3), the first term on the righthand side is asymptotically negligible relative to the other two terms. If $\mathbb{E} X_{ig}^4 < \infty$, then the variance estimators are also asymptotically normal. For the homoskedastic variance estimator, this is a straightforward application of the central limit theorem:

$$\begin{aligned} \sqrt{n}(\hat{\sigma}_h^2 - \sigma^2) &= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} ((X_{gi} - \bar{X}_n)^2 - \sigma^2) \\ &= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} ((X_{gi} - \mu)^2 - \sigma^2) - \sqrt{n}(\bar{X}_n - \mu)^2 \\ &\Rightarrow \mathcal{N}(0, \sigma^4(\kappa - 1)) \end{aligned}$$

where κ is the kurtosis of X_{ig} . For the cluster-robust variance estimator, a computationally convenient

way to understand the asymptotic behaviour is to consider the cluster-averages $\tilde{X}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} X_{ig}$ as the observations rather than the X_{ig} . Using the relationships between central moments and cumulants, as well as the independence of the X_{ig} , we have that

$$\begin{aligned}
\sqrt{n}(\hat{\sigma}_c^2 - \sigma^2) &= \sqrt{n} \left(\frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{gi} - \bar{X}_n) \right)^2 - \sigma^2 \right) \\
&= \sqrt{n} n_g \left(\frac{1}{G} \sum_{g=1}^G \left((\tilde{X}_g - \mu)^2 - \frac{\sigma^2}{n_g} \right) - (\bar{X}_n - \mu)^2 \right) \\
&= n_g^{3/2} \frac{1}{\sqrt{G}} \sum_{g=1}^G \left((\tilde{X}_g - \mu)^2 - \frac{\sigma^2}{n_g} \right) - n_g \sqrt{n} (\bar{X}_n - \mu)^2 \\
&\Rightarrow \mathcal{N}(0, \sigma^4(\kappa - 1) + 2\sigma^4(n_g - 1)).
\end{aligned} \tag{2.4}$$

The variance of the cluster-robust variance estimator is inflated relative to the homoskedastic version by a factor of $2\sigma^4(n_g - 1)$. If we consider the standardised difference between the estimators $\sqrt{n}(\hat{\sigma}_h^2 - \hat{\sigma}_c^2)$, we obtain that

$$\begin{aligned}
\sqrt{n}(\hat{\sigma}_c^2 - \hat{\sigma}_h^2) &= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) - \sqrt{n}(n_g - 1)(\bar{X}_n - \mu)^2 \\
&= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) + o_P(1).
\end{aligned}$$

When the observations X_{ig} are i.i.d., these terms are mean-zero, and thus do not alter the conclusion that $\hat{\sigma}_c^2 - \hat{\sigma}_h^2 \xrightarrow{P} 0$, however they do contribute to the variance:

$$\begin{aligned}
\text{Var} \left(\frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) \right) &= \frac{1}{G} \sum_{g=1}^G \frac{1}{n_g} \text{Var} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) \right) \\
&= \frac{1}{n_g} \text{Var} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) \right) \\
&= \frac{1}{n_g} \mathbb{E} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_j - \mu) \right)^2
\end{aligned}$$

The sum has $n_g(n_g - 1)$ terms of the form $(X_{ig} - \mu)(X_j - \mu)$. Expanding the square, we end up with $n_g(n_g - 1)$ terms $(X_{ig} - \mu)^2(X_j - \mu)^2$ from squaring each entry, as well as an additional identical $n_g(n_g - 1)$ terms from the product of $(X_{ig} - \mu)(X_j - \mu)$ with $(X_j - \mu)(X_{ig} - \mu)$. All other products that come out of the sum are mean zero: $\mathbb{E}((X_{ig} - \mu)(X_j - \mu)(X_l - \mu)(X_m - \mu)) = 0$ unless $i = l$ and $j = m$ or $i = m$ and $j = l$. Thus, the variance is made up of $2n_g(n_g - 1)$ terms, each equal to $\mathbb{E}(X_{ig} - \mu)^2(X_j - \mu)^2 = \sigma^4$. In this way, we can understand the asymptotic variance in (2.4) as a sum of two variances. $\sigma^4(\kappa - 1)$ is the asymptotic variance of $\hat{\sigma}_h^2$, and is related to the fourth-moments of the X_{ig} . The $2\sigma^4(n_g - 1)$ term is the variance of the additional terms that must be included when achieving robustness with respect to cluster dependence.

Now, consider again the expansion (2.3). For the distribution of $\sqrt{n}(\frac{1}{n}W_a - \xi^2)$, we need the joint limit distribution of $\bar{X}_n$ and $\hat{\sigma}_a^2$. Defining γ as the skewness of X_{ig} and applying the delta-method to (2.3), we

have

$$\sqrt{n} \left(\frac{1}{n} W_a - \xi^2 \right) \Rightarrow \mathcal{N}(0, \Sigma_a) \quad (2.5)$$

where

$$\Sigma_h = \xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4) \quad (2.6)$$

$$\Sigma_c = \Sigma_h + 2(n_g - 1)\xi^2. \quad (2.7)$$

In the application of the delta-method to (2.3) we recognise that the first term is $O_P(1/n)$: it is the Wald statistic to test the null hypothesis $H_0 : \mathbb{E} X_{ig} = \mu$. Thus, it is the second two terms that are important for the asymptotic distribution of $\frac{1}{n} W_a$. We then have

$$\frac{1}{n} W_a - \xi^2 = \frac{2\Delta(\bar{X}_n - \mu)}{\hat{\sigma}_a^2} + \frac{\Delta^2}{\hat{\sigma}_a^2} - \frac{\Delta^2}{\sigma^2} + O_P(1/n) \quad (2.8)$$

$$= \frac{2\Delta}{\sigma^2}(\bar{X}_n - \mu) - \frac{\Delta^2}{\sigma^4}(\hat{\sigma}_a^2 - \sigma^2) + O_P(1/n) + o_P(|\hat{\sigma}_a^2 - \sigma^2|) \quad (2.9)$$

This stochastic expansion is valid whenever $\sigma^2 \geq c > 0$ for some c , and will be useful to recall later when looking at the general case where the convergence rate for $\hat{\sigma}_a^2$ and $\bar{X}_n$ might be different. We can see from (2.9) that characterizing the asymptotic distribution of the Wald test statistics under a fixed alternative is equivalent to characterizing the asymptotic distribution of a linear combination of the estimator from the numerator, $\bar{X}_n$, and the estimator of the variance, $\hat{\sigma}_a^2$. Since we have already shown that the variance estimators in this sample have have different first-order behaviour, the limit distribution of W_a (properly centered and normalized) will vary depending on which variance estimator is used.

The computationally convenient expansion of $\hat{\sigma}_c^2$ in terms of the cluster averages $\tilde{X}_g$ gives one interpretation of the difference between the two test statistics: the cluster-robust variance estimator is effectively using only G observations for estimating the variance, rather than n , thereby inflating the variance by a factor of approximately $n/G = n_g$. The natural interpretation based on looking at the difference $\hat{\sigma}_c^2 - \hat{\sigma}_h^2$ is that the robustness provided by $\hat{\sigma}_c^2$ leads to additional terms in the variance estimator. These terms are mean-zero, and therefore drop out while applying Slutsky's theorem under local alternatives, however they contribute to the variance of the test statistic under an alternative which is a fixed distance from the null.

We now connect the asymptotic distribution of the test statistics to power. Let $C_{1-\alpha}^2$ be the upper α quantile of a χ_1^2 random variable. Local alternatives give a (local) asymptotic approximation to power: under the sequence of local alternatives $\mu_n = \mu_0 + \delta/\sqrt{n}$, we have that

$$P(W_a > C_{1-\alpha}^2) \rightarrow 1 - F_{\chi_1^2(\delta^2/\sigma^2)}(C_{1-\alpha}^2), \quad a \in \{h, c\}. \quad (2.10)$$

The limiting power function is that of a non-central chi-square distribution which is invariant to the choice of variance estimator. The asymptotic power comparisons under local alternatives do not distinguish between Wald tests where different consistent variance estimators are used.

One implication of (2.5), (2.6), and (2.7) is that under fixed alternatives the test statistics have different asymptotic distributions. This implies that we can design a sequence of tests which leads to non-degenerate power under the fixed alternative $\mathbb{E} X_{ig} = \mu = \mu_0 + \Delta$. Let C_n^a denote a sequence of critical values for each

test statistic. Using the asymptotic distribution in (2.5), we have that

$$\begin{aligned} P(W_a > C_n^a) &= P\left(\left(\frac{1}{n}W_a - \xi^2\right) > \frac{C_n^a}{n} - \xi^2\right) \\ &= P\left(\Sigma_a^{-1/2}\sqrt{n}\left(\frac{1}{n}W_a - \xi^2\right) > \frac{C_n^a}{\sqrt{n}\Sigma_a} - \xi^2\sqrt{\frac{n}{\Sigma_a}}\right). \end{aligned} \quad (2.11)$$

This suggests a choice for C_n^a . If we choose

$$C_n^a = n\left(\xi^2 - t\sqrt{\frac{\Sigma_a}{n}}\right) \quad (2.12)$$

then we have that under the alternative $\mathbb{E} X_{ig} = \mu_0 + \Delta$,

$$P(W_a > C_n^a) = P\left(\Sigma_a^{-1/2}\sqrt{n}\left(\frac{1}{n}W_a - \xi^2\right) > -t\right) \rightarrow 1 - \Phi(t) \quad (2.13)$$

Part of the appeal of local power analysis is that asymptotically power is in $(0, 1)$. Under a fixed alternative, this sequence of critical values will lead to such nondegenerate power in the limit. If we choose $t = C_\beta$, then $P(W_a > C_n^a) \rightarrow 1 - \beta$, noting that the sequence of critical values will be different for each test statistic to achieve the same power. The chosen sequence C_n^a is local to the (scaled) non-centrality parameter $n\xi^2$.

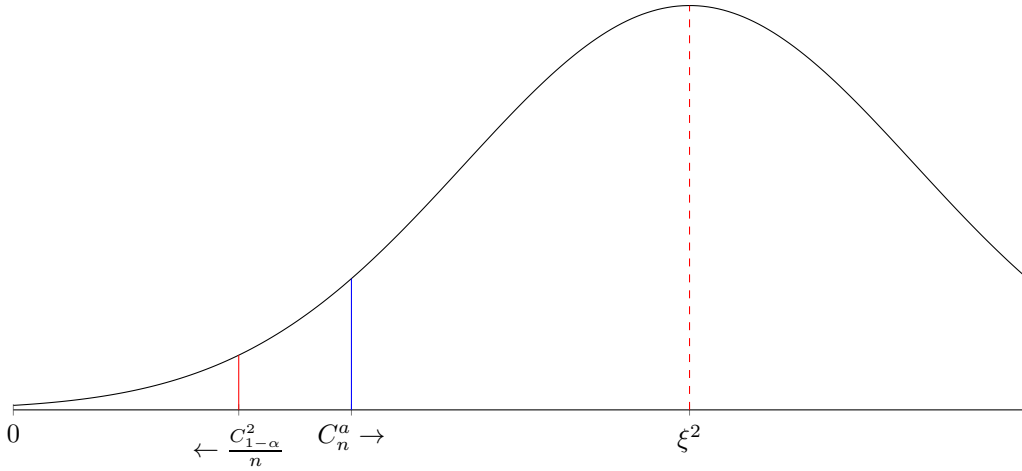


Figure 1: The distribution of $\frac{1}{n}W_a$ will concentrate around the non-centrality parameter ξ^2 , while the critical value for the test $C_{1-\alpha}^2/n$ converges to 0. The chosen sequence C_n^a will converge to the non-centrality parameter at the correct rate so that the power of this sequence of tests is non-degenerate asymptotically.

In Figure 1, we lay out the relationship between the non-centrality parameter, C_n^a , and $C_{1-\alpha}^2/n$. We want to gain insight into asymptotic behaviour of the exact power in (2.11), and therefore choose C_n^a converging to ξ^2 . Note that any sequence such that $\sqrt{n}(C_n - n\xi^2)$ tends to a constant will have this property, and the choice given is for convenience. A similar asymptotic power approximation was proposed in Bentkus et al. (2007) in the case of the 1-sample t-test. In that paper they were primarily concerned with variations in behaviour of the t-test under a fixed alternative when the observations had possibly fewer than 4 finite moments.

For our two test statistics we have thus considered, the considerations on power lead to

$$C_n^h = n \left(\xi^2 - C_\beta \sqrt{\frac{\xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4)}{n}} \right)$$

$$C_n^c = n \left(\xi^2 - C_\beta \sqrt{\frac{\xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4) + 2(n_g - 1)\xi^2}{n}} \right).$$

Along these sequences, we have that

$$P(W_n^h > C_n^h) \rightarrow 1 - \beta$$

$$P(W_n^c > C_n^c) \rightarrow 1 - \beta.$$

Because the two test statistics have different asymptotic distributions under a fixed-alternative, difference sequences of critical values are required to achieve the same power. Thus, this analysis has lead to a sequence of testing problems which is sensitive to the choice of variance estimator. Since we have not discussed Type-I errors along this sequence, it is not clear how this should imply a preference-ordering between the two test statistics. The main conclusion to be drawn at this stage is that a necessary condition for the first-order properties of a sequence of tests to depend on the choice of variance estimator is that size converges to 0.

The given choice of critical values implies that the size of the sequence of tests converges to 0. This feature is reminiscent of Bahadur asymptotic relative efficiency (Bahadur (1960a), Bahadur (1960b), and Bahadur (1967); see also Serfling (2009)). Applying that setup to our simple example, the comparison is based on the rate at which the p-values converge to 0 under a fixed alternative. Let $F_{0,n}^a$ be the CDF of the test statistics W_n^a under the null hypothesis. The p-values are then $1 - F_{0,n}^a(W_n^a)$. The Bahadur slope of the test $c(\Delta)$ is the rate at which the p-values converge to 0,

$$\lim_{n \rightarrow \infty} -\frac{2 \log(1 - F_{0,n}^a(W_n^a))}{n} = c_a(\Delta).$$

In that setup, $c_a(\Delta) > 0$ when W_n^a satisfies a large-deviation property under the null hypothesis. If such a property holds (as is the case, for example, when the X_{ig} are normally distributed) then we can compare the functions $c_a(\Delta)$ to compare the test statistics, with larger values being better as they reflect a faster convergence rate of the p-values to 0. After covering the general case of asymptotic power for Wald test statistics under fixed-alternatives, we return to how our approach augments the Bahadur approach for comparing test statistics in Section 4.

3 Fixed-Alternative Asymptotics

The previous section motivates the following derivation of the asymptotic distribution of Wald test statistics under fixed alternatives. In this section we introduce the general setup for deriving the asymptotic distribution of Wald test statistics under fixed alternatives for asymptotically linear estimators. We first give sufficient conditions for when the test statistics are asymptotically normal under a fixed alternative, and then use this first-order approximation to motivate an alternative approach to power calculations.

3.1 Test Statistics under Fixed-Alternatives

We focus our attention on parameters which are identified by estimating equations:

$$\mathbb{E} \psi(X_{ig}, \beta) = 0 \quad (3.1)$$

where $X_{ig} \in \mathcal{X}$, $\beta \in \mathcal{B} \subset \mathbb{R}^p$, and $\psi : \mathcal{X} \times \mathcal{B} \rightarrow \mathbb{R}^q$, with $q \geq p$. Our focus will be on tests of a linear hypothesis. Define the parameter $\theta := \ell' \beta$ for a fixed vector ℓ . Tests of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ are commonly performed in applied work using the Wald statistic:

$$W_n := n \frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{V}_n \ell} \quad (3.2)$$

where $\hat{V}_n$ is a suitable variance estimator of a constant matrix V_n such that $V_n^{-1/2} \sqrt{n}(\hat{\beta} - \beta) \Rightarrow \mathcal{N}(0, I_p)$. Under standard regularity conditions, W_n is asymptotically χ_k^2 .

When $\theta \neq \theta_0$, W_n will diverge off to ∞ , in the sense that for all $C > 0$, $P(W_n < C) \rightarrow 0$. This is clearly not useful for characterising the behaviour of W_n in large samples, and therefore we will consider suitable centring and (re)scaling. Our first observation towards this goal is that under the same conditions providing for W_n to be asymptotically χ_k^2 , for $\Delta = \ell' \beta - \theta_0$, we have that:

$$n \frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{V}_n \ell} - \frac{\Delta^2}{\ell' V_n \ell} \xrightarrow{P} 0. \quad (3.3)$$

Under a fixed alternative we can view $\frac{1}{n} W_n$ as a consistent estimator of the noncentrality parameter $\xi_n^2 := \frac{\Delta^2}{\ell' V_n \ell}$. In the GMM setting we consider here, $\frac{1}{n} W_n$ is a function of three random components: $\hat{\beta}$, $\hat{\Omega}_n$, and $\hat{Q}_n$, with $\hat{V}_n = (\hat{Q}_n' \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1}$ taking the usual sandwich form. Under regularity conditions on $\hat{Q}_n$ and $\hat{\Omega}_n$ and their probability limits, we will be able to write the test statistic as a linear functional of three terms under a fixed alternative,

$$\frac{1}{n} W_n - \xi_n^2 \approx p'_{1n}(\hat{\beta} - \beta) + \text{tr}(P_{1n}(\hat{Q}_n - Q_n)) + \text{tr}(P_{2n}(\hat{\Omega}_n - \Omega_n)). \quad (3.4)$$

If we can characterise the behaviour of Using the delta-method, we can then derive the normalising sequence Ξ_n and possible bias term B_n such that

$$\Xi_n^{-1/2} \left(\frac{1}{n} W_n - \xi_n^2 - B_n \right) \Rightarrow \mathcal{N}(0, 1). \quad (3.5)$$

We can then conclude that for critical values of the form

$$C_n = n(\xi_n^2 + B_n) - \frac{n C_\beta}{\sqrt{\Xi}} \quad (3.6)$$

that $P(W_n > C_n) \rightarrow 1 - \beta$. In the following subsections we will give explicit expressions for p_{1n} , P_{1n} , P_{2n} , B_n , and Ξ_n in the case that the Wald statistic is formed using a plug-in estimator of V_n using the efficient GMM estimator $\hat{\beta}$. In the example of the t-test, $B_n = 0$ and Ξ_n is of the order $1/\sqrt{n}$, but we will discuss a few examples where the more general case is of interest. Other estimators for $\hat{V}_n$ are also of interest theoretically, but are less common in practise. One exception is an estimator using $\tilde{\beta}_0$, an estimate of β under the restriction $\ell' \beta = \theta_0$. We do not consider this case in general, however we will discuss the

particular comparison of the score test for a normal mean below.

3.2 Assumptions

In this section we lay out our high-level conditions which allow us to derive the asymptotic distribution of Wald test statistics under fixed alternatives. As cluster-robust inference is one of our main applications, the first assumption places restrictions on the sizes of the clusters.

Assumption 1. (*Cluster dependence*) Denote X_{ig} as observation i in cluster g . Observations are independent across clusters, and we allow for arbitrary cluster dependence within cluster g , where there are G clusters. If cluster g has n_g observations, the overall sample size is $n = \sum_{g=1}^G n_g$. We require $G \rightarrow \infty$ as $n \rightarrow \infty$, and for some $r \in [2, \infty)$ and $C < \infty$, we have that

$$\frac{\left(\sum_{g=1}^G n_g^{2r}\right)^{2/r}}{n} \leq C, \quad \lim_{n \rightarrow \infty} \max_{g \in \{1, \dots, G\}} \left\{ \frac{n_g^4}{n} \right\} = 0. \quad (3.7)$$

This assumption is from Hansen and Lee (2019) and ensures that clusters cannot be too heterogeneous in size or grow too quickly with n for a central limit theorem to hold. The assumption is also similar to Assumption 3 in Djogbenou, MacKinnon, and Nielsen (2019). Here we use the version in Assumption 3 from Hansen and Lee (2019) paper, as we will be characterising the asymptotic behaviour of estimators of second moments. In this case, the estimator of the variance cannot be written as a sum over i , but is rather written as a sum over the clusters g , which requires the second part of (3.7) to have n_g^4 instead of n_g^2 . The constant r will be connected to the number of finite moments required below. Our conditions will imply that $\hat{\beta}$, $\hat{Q}_n$, and $\hat{\Omega}_n$ are asymptotically linear, and these influence functions will be required to have at least r moments, which at a minimum will require that $\mathbb{E} \|\psi(X_{ig}, \beta)\|^4 < \infty$ so that the variance estimator is asymptotically normal; if the clusters are particularly large or heterogeneous in size, this will lead to stronger restrictions on the moments in the assumptions below. In several of our examples below, it is of sufficient interest to set $n_g = 1$ and $G = n$ and this assumption is trivially satisfied. The assumption is also satisfied whenever n_g is fixed for all sample sizes and n is sufficiently large. It also applies when the clusters are growing, e.g. $n_g = n^\alpha$ for some $\alpha < (r - 2)/2(r - 1)$. For more discussion of the restrictions Assumption 1 places on the cluster sizes, see Hansen and Lee (2019) and Djogbenou et al. (2019).

The next assumption confirms the definition of β , $\hat{\beta}$, and provides for W_n leading to valid inference when $\beta = \beta_0$. Before doing this we will define Q_n and Ω_n as the standard components of the asymptotic variance of $\hat{\beta}$ estimated using efficient GMM. Let $\dot{\psi}_{ig}(\beta) = \frac{\partial}{\partial \beta} \mathbb{E} \psi(X_{ig}, \beta)$ and $\tilde{\psi}_g(X_g, \beta) := \sum_{i=1}^{n_g} \psi_{ig}(X_{ig}, \beta)$. Then, we define

$$Q_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \dot{\psi}_{ig}(\beta) \quad (3.8)$$

$$\Omega_n := \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' \quad (3.9)$$

We will use plug-in estimators of Q_n and Ω_n , so that

$$\hat{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \hat{\beta}) \quad (3.10)$$

$$\hat{\Omega}_n := \frac{1}{n} \sum_{g=1}^G \tilde{\psi}(X_g, \hat{\beta}) \tilde{\psi}(X_g, \hat{\beta})' \quad (3.11)$$

for some function $\hat{\psi}$.

Assumption 2. We assume that there exists a unique $\beta \in \mathcal{B} \subset R^p$ such that $\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \psi(X_{ig}, \beta) = 0$. We further assume that $\mathbb{E} \psi(X_{ig}, b)$, $\mathbb{E} \hat{\psi}_{lm}(X_{ig}, b)$ and $\mathbb{E} \psi_l(X_{ig}, b) \psi_m(X_{ig})$ are continuously differentiable in b for all indices l, m , for all b in an open neighbourhood N_β which contains β . We will denote $\frac{\partial}{\partial \beta} \mathbb{E} \psi(X_{ig}, b) =: \dot{\psi}_{ig}(b)$. We assume that

$$\lim_{M \rightarrow \infty} \sup_{i,g} \sup_{\beta \in N_\beta} \mathbb{E} \|\psi(X_{ig}, \beta)\|^{2r} \mathbf{1}_{[\|\psi(X_{ig}, \beta)\| > M]} = 0 \quad (3.12)$$

$$\lim_{M \rightarrow \infty} \sup_{i,g} \sup_{\beta \in N_\beta} \mathbb{E} \|\hat{\psi}(X_{ig}, \beta)\|^r \mathbf{1}_{[\|\hat{\psi}(X_{ig}, \beta)\| > M]} = 0 \quad (3.13)$$

where r satisfies Assumption 1. The matrices Q_n and Ω_n are full-rank, and the minimum eigenvalue $\lambda_{\min}(\Omega_n)$ satisfies

$$\lambda_{\min}(\Omega_n) \geq c > 0$$

for some c . The estimator $\hat{\beta}$ is an efficient GMM estimator which satisfies

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\beta}) = o_P(\|\Omega_n\|^{1/2}) \quad (3.14)$$

$$(Q'_n \Omega_n^{-1} Q_n)^{1/2} \sqrt{n}(\hat{\beta} - \beta) = (Q'_n \Omega_n^{-1} Q_n)^{-1/2} \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} Q'_n \Omega_n^{-1} \psi(X_{ig}, \beta) + o_P(1) \Rightarrow \mathcal{N}(0, I_p). \quad (3.15)$$

These are high-level assumptions that assure the Wald statistic provides valid inference under the null. We also require additional smoothness of $\mathbb{E} \psi(X_{ig}, b)$, $\mathbb{E} \psi(X_{ig}, b) \psi(X_{ig}, b)'$, and $\mathbb{E} \hat{\psi}(X_{ig}, b)$. This allows use to use the delta-method when deriving the asymptotic behaviour of the variance estimators. The moment conditions are stronger than required for $W_n \Rightarrow \chi_k^2$, however the Lindeberg conditions will be sufficient for W_n to have a limit distribution under the alternative. The assumptions as stated serve the purpose of clarifying the environment: inference settings where asymptotically normal inference is correct in large samples. We now discuss verification of Assumption 2 in a few examples.

Example 1 (*t-statistic*). For the t-statistic with X_{ig} independent across i and g , $Q_n = 1$ and $\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} (\sum_{i=1}^{n_g} (X_{ig} - \mu))^2 = \text{Var}(X_{ig}) = \sigma^2$. Thus, the main substantive assumptions in Assumption 2 are that $\mathbb{E} X_{ig}^4 < \infty$ and there exists some c such that $\sigma^2 \geq c > 0$. In this case we can apply a classic central limit theorem to obtain that $\frac{1}{\sigma} \sqrt{n}(\bar{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{1}{\sigma} (X_{ig} - \mu) \Rightarrow \mathcal{N}(0, 1)$.

Example 2 (*Efficient Smooth-GMM*). If $\psi(X_{ig}, \beta)$ is twice-continuously differentiable in β , $\mathbb{E} \frac{\partial}{\partial \beta} \psi(X_{ig}, \beta)$ has full-rank, $\mathbb{E} \|\psi(X_{ig}, \beta)\|^2 < \infty$, and the data are i.i.d., the efficient-GMM estimator $\hat{\beta}$ will generally satisfy (3.15) with

$$Q_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \frac{\partial}{\partial \beta} \psi(X_{ig}, \beta) = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \dot{\psi}_{ig}(\beta) \quad (3.16)$$

$$\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}(X_g, \beta) \tilde{\psi}(X_g, \beta)' \quad (3.17)$$

For verifying (3.15), standard conditions involve assuming $\mathcal{B}$ is compact, Ω_n is of full rank with minimum eigenvalue bounded away from zero, Q_n is full rank, and $\sup_{i,g} \sup_{\beta \in \mathcal{B}} \dot{\psi}_{ig}(\beta) \leq C$ for some $C < \infty$. See Hansen and Lee (2019) for a full list of conditions for validity of Wald tests in a GMM setting with clustering, as well as Newey and McFadden (1994), Theorem 3.2 for a discussion of the i.i.d. case when $G = n$ and $n_g = 1$. In the smooth moment case, (3.14) will be satisfied as the first-order condition will typically hold exactly, and therefore $\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\beta}) = 0$.

The main assumptions required here that are not required for the asymptotic validity of the Wald test statistic are the existence of the second moments for the derivative matrix and fourth moments for the moment conditions; in other words, we will assume that there exists an $\epsilon > 0$ such that

$$\begin{aligned} \sup_{i,g} \sup_{\beta \in \mathcal{B}} \mathbb{E} \|\psi(X_{ig}, \beta)\|^{2r+\epsilon} &< \infty \\ \sup_{i,g} \sup_{\beta \in \mathcal{B}} \mathbb{E} \left\| \frac{\partial}{\partial \beta} \psi(X_{ig}, \beta) \right\|^{r+\epsilon} &< \infty \end{aligned}$$

where $r \geq 2$ is as in Assumption 1.

Example 3 (*Cluster-Robust Regression*). Consider the standard linear regression model:

$$y_{ig} = x'_{ig} \beta + \varepsilon_{ig} \quad (3.18)$$

Here, $\psi(X_{ig}, \beta) = x_{ig}(y_{ig} - x'_{ig} \beta) = x_{ig} \varepsilon_{ig}$. In this case, we have

$$Q_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} x_{ig} x'_{ig} \quad (3.19)$$

$$\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \left[\left(\sum_{i=1}^{n_g} x_{ig} \varepsilon_{ig} \right) \left(\sum_{i=1}^{n_g} x_{ig} \varepsilon_{ig} \right)' \right]. \quad (3.20)$$

The moment conditions will be satisfied if require that for some $\epsilon > 0$, $\mathbb{E} \|x_{ig}\|^{2r+\epsilon} < \infty$ and $\mathbb{E} y_{ig}^{2r+\epsilon} < \infty$. Standard assumptions on the design imply that $\lambda_{\min}(\Omega_n)$ is bounded away from zero and Q_n is full rank. Since (3.14) is satisfied exactly for the least-squares estimator $\hat{\beta}$, (3.15) will hold.

Example 4 (Quantile Regression). Consider the quantile regression model:

$$y_{ig} = x'_{ig}\beta(\tau) + \varepsilon_{ig}(\tau). \quad (3.21)$$

We will assume in this example that $G = n$ and $n_g = 1$, with i.i.d. (y_{ig}, x_{ig}) . In this setting, $\psi(X_{ig}, \beta(\tau)) = x_{ig} \left(\mathbb{1}_{[y_{ig} - x'_{ig}\beta(\tau) < 0]} - \tau \right)$, and

$$\dot{\psi}_{ig}(\beta(\tau)) = \mathbb{E} x_{ig} x'_{ig} f_\tau(\varepsilon_{ig}(\tau) | x_{ig}) \quad (3.22)$$

where $f_\tau(\cdot | x_{ig})$ is the conditional density of $\varepsilon_{ig}(\tau)$ given x_{ig} . The expressions for Ω_n and Q_n are then

$$\begin{aligned} Q_n &= \dot{\psi}_{ig}(\beta(\tau)) \\ &= \mathbb{E} x_{ig} x'_{ig} f_\tau(0 | x_{ig}) \\ \Omega_n &= \mathbb{E} x_{ig} x'_{ig} \left(\mathbb{1}_{[y_{ig} - x'_{ig}\beta(\tau) < 0]} - \tau \right)^2, \end{aligned}$$

When $\sup_{ig} \sup_u f_\tau(u | x_{ig}) \leq C < \infty$ and $\mathbb{E} \|x_{ig}\|^{4+\epsilon} < \infty$ then (3.15) is satisfied.

In Example 4, the function $\dot{\psi}_{ig}$ depends on the unknown density $f_\tau(u | x_{ig})$. This case motivates our next assumption, and is useful to verify in proving asymptotic normality of the kernel estimators $\hat{Q}_n$ typically used in the quantile regression literature.

Assumption 3. *We assume that*

$$\begin{aligned} \|\hat{\Omega}_n - \Omega_n\| &\xrightarrow{P} 0 \\ \|\hat{Q}_n - Q_n\| &\xrightarrow{P} 0 \end{aligned}$$

where here $\|\cdot\|$ is the spectral norm. We further assume that for any r_{1n} and r_{2n} such that

$$r_{1n} \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\beta) - \mathbb{E} \hat{\psi}_{ig}(\beta) \right) = O_P(1), \quad r_{1n}(\hat{\beta} - \beta) = O_P(1) \quad (3.23)$$

$$r_{2n} \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' - \Omega_n \right) = O_P(1), \quad r_{2n}(\hat{\beta} - \beta) = O_P(1) \quad (3.24)$$

we have that

$$r_{1n} \left(\hat{Q}_n - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \hat{\psi}_{ig}(\hat{\beta}) - \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\beta) - \mathbb{E} \hat{\psi}_{ig}(\beta) \right) \right) = o_P(1) \quad (3.25)$$

$$r_{2n} \left(\hat{\Omega}_n - \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \hat{\beta}) \tilde{\psi}_g(X_g, \hat{\beta})' - \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' - \Omega_n \right) \right) = o_P(1). \quad (3.26)$$

In characterising the asymptotic distributions of $\hat{Q}_n$ and $\hat{\Omega}_n$, we cannot rule out that the estimation of β impacts the asymptotic behaviour of these matrices, which take $\hat{\beta}$ as an input. This assumption will imply that the estimation of β only enters in the first-order asymptotic behaviour of $\hat{Q}_n$ and $\hat{\Omega}_n$ through

$\mathbb{E} \hat{\psi}_{ig}(\hat{\beta})$ and $\mathbb{E} \tilde{\psi}_g(X_g, \hat{\beta}) \tilde{\psi}(X_g, \hat{\beta})'$, which can effectively be handled using the delta-method. Furthermore, the additional noise due to estimating β must go to zero at a faster rate than the estimation error in estimating Q_n or Ω_n if β was known. This condition is typically satisfied when the $\hat{\psi}$ and ψ functions are sufficiently smooth in β to satisfy a stochastic-equicontinuity condition. Differentiable functions, Lipschitz functions, and more generally Donsker classes will generally satisfy these conditions. In the case of the t-test and linear models (Examples 1 and 3), these requirements are almost trivially satisfied. The cases of GMM and quantile regression deserves a bit more attention.

Example 1, continued (*t-statistic*). As long as $\mathbb{E} X_{ig}^4 < \infty$, we have that $\sqrt{n}(\hat{\sigma}_a^2 - \sigma^2) = O_P(1)$ for both the homoskedastic and cluster-robust variance estimators. Furthermore, we have that for $\hat{\sigma}_c^2$,

$$\begin{aligned} \mathbb{E} \hat{\sigma}_c^2 &= \sigma^2 - \frac{n_g}{n} \sigma^2 \\ \sqrt{n} \left(\hat{\sigma}_c^2 - \mathbb{E} \hat{\sigma}_c^2 - \left(\frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} X_{ig} - \mu \right)^2 - \sigma^2 \right) \right) \\ &= \sqrt{n} \left(\frac{n_g}{n} \sigma^2 - n_g (\bar{X}_n - \mu)^2 \right) \\ &\xrightarrow{P} 0 \end{aligned}$$

with a similar argument showing that $\hat{\sigma}_h^2$ also satisfies (3.26).

Example 2, continued (*Efficient Smooth-GMM*). In the case of smooth GMM settings when $\psi(X_{ig}, \beta)$ is twice-differentiable, then Assumption 3 will be satisfied.

Assumption 3^{gmm} (*Smoothness assumption for GMM*). Let there exist a neighbourhood N_β of β such that $\psi(X_{ig}, \beta)$ is twice-continuously differentiable on N_β for all X_{ig} , and let $\psi_l(X_{ig}, \beta)$ denote the l^{th} element of the function $\psi_l(X_{ig}, \beta)$. We assume that there exists a C such that

$$\begin{aligned} \sup_{i,g,l,m} \sup_{b \in N_\beta} \mathbb{E} \left\| \left(\frac{\partial^2}{\partial b \partial b'} \psi_l(X_{ig}, b) \right) \right\|^{r+\epsilon} &\leq C < \infty \\ \sup_{i,g} \sup_{b \in N_\beta} \mathbb{E} \left\| \left(\frac{\partial}{\partial b} \psi(X_{ig}, b) \right)' \psi(X_{ig}, b) \right\|^{r+\epsilon} &\leq C < \infty \end{aligned}$$

for all $b \in N_\beta$.

Under this condition, Assumption 3 will be satisfied.

Lemma 1. Under Assumption 3^{gmm}, we have that

$$\hat{Q}_n - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \hat{\psi}_{ig}(\hat{\beta}) - \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\beta) - Q_n \right) = o_P(\|\hat{\beta} - \beta\|) \quad (3.27)$$

$$\hat{\Omega}_n - \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \hat{\beta}) \tilde{\psi}(X_g, \hat{\beta})' - \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \beta) \tilde{\psi}(X_g, \beta)' - \Omega_n \right) = o_P(\|\hat{\beta} - \beta\|) \quad (3.28)$$

An easy way to verify the second part of Assumption 3^{gmm} is to check that $\sup_{i,g} \sup_b \mathbb{E} \left\| \frac{\partial}{\partial b} \psi(X_{ig}, b) \right\|^{2r+\epsilon} < \infty$. Assumption 3^{gmm} can then be used to prove Lemma 1 by providing a Lipchitz-type bound on the empirical process. In the context of Assumption 3, we can choose the rates r_{1n}, r_{2n} to be of order $\frac{\sqrt{n}}{\sqrt{\ell' V_n \ell}}$.

Example 3, continued (*Cluster-Robust Regression*). In this case, (3.25) holds trivially as it does not depend on β . For $\hat{\Omega}_n$, the setup is similar to the t-statistic, however it is perhaps more straightforward to observe that Assumption 3^{gmm}: if $\sup_{i,g} \mathbb{E} \|x_{ig}\|^{2r+\epsilon} < \infty$ then this assumption is satisfied, which we have already assumed to verify Assumption 2.

Example 4, continued (*Quantile Regression*). In the case of quantile regression, recalling that we have assumed the observations are i.i.d., we have to estimate:

$$Q_n = \mathbb{E} x_{ig} x'_{ig} f_\tau(0|x_{ig}) \quad (3.29)$$

A standard estimator is a kernel estimator. Let $K(u)$ be a symmetric function of bounded variation such that $\int_{\mathbb{R}} uK(u)du = 0$ and $\int_{\mathbb{R}} u^2K(u)du = 1$. A kernel estimator of Q_n is given by

$$\hat{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{y_{ig} - x'_{ig} \hat{\beta}_\tau}{h_n}\right) \quad (3.30)$$

where h_n is a bandwidth. Thus, we have that, in the notation we have used thus far, $\hat{\Psi}_{ig}(\hat{\beta}_\tau) = x_{ig} x'_{ig} h_n^{-1} K((y_{ig} - x'_{ig} \hat{\beta}_\tau)/h_n)$. Assumption 3 is then satisfied with the following assumptions on K and the density of the error $\varepsilon_i(\tau)$:

Assumption 3^{reg} (*Conditional density estimation for quantile regression*).

1. The kernel function K is symmetric, of bounded variation, and normalized such that:

$$\int_{\mathbb{R}} uK(u)du = 0, \quad \int_{\mathbb{R}} u^2K(u)du = 1 \quad (3.31)$$

2. There exist functions $G_j(x_{ig})$ such that for all x_{ig} , $G_j(x_{ig}) \geq |f^{(j)}(u|x_{ig})|$, uniformly in u , $j \in \{0, 1, 2\}$. Furthermore, G_j also satisfy, for some $\delta_j > 0$, $\mathbb{E}(G_0(x_{ig})\|x_{ig}\|^{4+\delta_0}) < \infty$, $\mathbb{E}(G_1(x_{ig})\|x_{ig}\|^{2+\delta_1})$, and $\mathbb{E}(G_2(x_{ig})\|x_{ig}\|^2) < \infty$.
3. $h_n = o(\log n/\sqrt{n})$.

The first assumption is satisfied by all kernel functions used in practice, such as the Gaussian, Epanechnikov, Uniform, Biweight, and Triweight kernels. The assumption on bounded variation implies that the function can only rise and fall finitely many times.

The assumption on the integrability of envelope functions G_j for the density and its derivatives is quite similar to assumptions used in Kato (2012) in proving asymptotic normality of the variance estimator when using the uniform kernel. Bounding the density and the first two derivatives is standard in the literature on kernel density estimation, and in the regression context due to the conditional nature of the density we must

impose additional restrictions on the regressors to ensure integrability of the effective envelope functions that are used in the bounds.

This bandwidth condition will allow the rate-optimal bandwidth, $h_n \propto n^{-1/5}$. It is slightly stronger than the bandwidth condition in Powell (1991), $n^2 h_n \rightarrow \infty$. In the estimation of $\mathbb{E} x_{ig} x'_{ig} f(0|x_{ig})$, we can decompose our kernel estimator into three components:

$$\hat{Q}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) \quad (3.32)$$

$$= \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) \quad (3.33)$$

$$+ \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) \quad (3.34)$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) \right) \quad (3.35)$$

$$+ \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right). \quad (3.36)$$

Considering the estimator this way, (3.35) will be asymptotically negligible at rate $o_P(1/\sqrt{nh_n})$, and this kind of result typically makes use of empirical process methods, and that is how we proceed here in proving Lemma 2 below. For (3.36), the delta-method applies. Finally, concerning (3.33) we have to assess whether additional centring is needed. The kernel density estimator requires recentring, since although (3.33) will be consistent for Q_n , its expectation differs from Q_n and thus we need to incorporate the bias term for asymptotic normality. Then, standard results on asymptotic normality of kernel density estimators give that, for any constant matrix A ,

$$\sqrt{nh_n} \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \text{tr} \left(A \left(x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) \right) \right) \right) \Rightarrow \mathcal{N}(0, V(A)) \quad (3.37)$$

for $V(A) = \mathbb{E}[(x'_{ig} A x_{ig})^2 f(0|x_{ig}) R_K]$ and $R_K = \int K(u)^2 du$ is the roughness of the kernel $K(\cdot)$. Assumption 3^{reg} implies that this is the only component of the asymptotic distribution of $\hat{Q}_n$ which is relevant for the first-order theory:

Lemma 2. *Under Assumption 3^{reg}, we have that*

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) \quad (3.38)$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) \right) = o_P(1/\sqrt{nh_n}) \quad (3.39)$$

$$\mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n}\right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{\varepsilon_{ig}(\tau)}{h_n}\right) = o_P(1/\sqrt{nh_n}) \quad (3.40)$$

Thus, the quantile regression model satisfies Assumption 3, where we can observe that the important rates to consider are $r_{1n} = \sqrt{nh_n}$ and $r_{2n} = \sqrt{n}$.

We are now ready to state our general result on the asymptotic distribution of W_n under fixed alternatives. First, so that the statement of theorem is focused on the key aspects, we define a sequence of constants B_n and Ξ_n . Recall that $V_n = (Q_n' \Omega_n^{-1} Q_n)^{-1}$. Define:

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c_n' \frac{\partial}{\partial \beta_1} \mathbb{E} \hat{\psi}(X_{ig}, \beta) b_n \\ \vdots \\ c_n' \frac{\partial}{\partial \beta_p} \mathbb{E} \hat{\psi}(X_{ig}, \beta) b_n \end{pmatrix} \quad (3.41)$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} c_n' \frac{\partial}{\partial \beta_1} \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' c_n \\ \vdots \\ c_n' \frac{\partial}{\partial \beta_p} \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' c_n \end{pmatrix} \quad (3.42)$$

$$a_n := -2 \frac{\Delta \ell}{\ell' V_n \ell} + J_n - S_n \quad (3.43)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2 V_n \ell \quad (3.44)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} \Omega_n^{-1} Q_n V_n \ell. \quad (3.45)$$

Then, we can define:

$$\tilde{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \beta) \quad (3.46)$$

$$B_n := \frac{\Delta^2}{\ell' (\mathbb{E} \tilde{Q}_n' \Omega^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} - \xi_n^2 \quad (3.47)$$

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a_n' (Q_n' \Omega_n^{-1} Q_n)^{-1} Q_n' \Omega_n^{-1} \psi(X_{ig}, \beta) \\ \sum_{i=1}^{n_g} c_n' (\hat{\psi}(X_{ig}, \beta) - \mathbb{E} \hat{\psi}(X_{ig}, \beta)) b_n \\ -c_n' (\tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' - \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)') c_n \end{pmatrix} \quad (3.48)$$

$$\Xi_n := \frac{1}{n^2} \sum_{g=1}^G \mathbb{E}(\mathbf{1}' Y_{gn})^2 \quad (3.49)$$

Notice that $n\sqrt{\Xi_n} \rightarrow \infty$, as we require the number of clusters G to be increasing, and $\Xi_n = 0$ whenever $\Delta = 0$.

Theorem 1. *Assume that Assumptions 1-3 are satisfied. Suppose that $\mathbf{1}' Y_{gn}$ is nondegenerate (implying $\Xi_n > 0$). Then, for the previously defined sequences Ξ_n , B_n , we have that*

$$\frac{\frac{1}{n} W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0, 1) \quad (3.50)$$

When standard plug-in estimators for Q_n are available in GMM settings, then there will be no bias term B_n . In the context of quantile regression, however, B_n will depend on the choice of bandwidth and kernel for the variance estimator. In general, however, different choices of the estimators for Q_n and Ω_n will lead to difference sequences B_n and Ξ_n . This highlights an important feature of these asymptotics: under fixed alternatives, the choice of variance estimator impacts the limit distribution of the test statistic.

We now discuss, and derive expressions for, B_n and Ξ_n in the context of our examples.

Example 1, continued (*t-test*). Focusing on the cluster-robust test statistic W_n^c , $\ell = 1$ and $V_n = \sigma^2$, so we have that $J_n = 0$, $S_n = 0$, $a_n = -\frac{2\Delta}{\sigma^2}$, $c_n = \frac{\Delta}{\sigma^2}$, ignoring b_n because $Q_n = 1$. Therefore,

$$B_n = 0$$

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} -\frac{2\Delta}{\sigma^2}(X_{ig} - \mu) \\ 0 \\ -\frac{\Delta^2}{\sigma^4} \left(\left(\sum_{i=1}^{n_g} (X_{ig} - \mu) \right)^2 - n_g \sigma^2 \right) \end{pmatrix}$$

Observe that in this case $\mathbf{1}'Y_{gn}$ is nondegenerate as long as X_{ig} is not a Binary random variable such that $(X_{ig} - \mu)^2$ is constant almost surely.³ Therefore, we obtain that

$$\Xi_n = \frac{1}{n} \xi_n^2 ((\kappa - 1)\xi_n^2 - 4\gamma\xi + 4) + 2(n_g - 1)\xi_n^2$$

Example 2, continued (*Efficient Smooth-GMM*). There is not much simplification to be found from the general case.

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c'_n \mathbb{E} \frac{\partial}{\partial \beta \partial \beta_1} \psi(X_{ig}, \beta) b_n \\ \vdots \\ c'_n \mathbb{E} \frac{\partial}{\partial \beta \partial \beta_p} \psi(X_{ig}, \beta) b_n \end{pmatrix} \quad (3.51)$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} \sum_{l=1}^q 2c_{nl}^2 \mathbb{E} \frac{\partial}{\partial \beta_1} \tilde{\psi}_{gl}(X_g, \beta) + 2 \sum_{l < m} c_{nm} c_{nl} \mathbb{E} \left[\tilde{\psi}_{gl}(X_g, \beta) \frac{\partial}{\partial \beta_1} \psi_{gm}(X_g, \beta) \right] \\ \vdots \\ \sum_{l=1}^q 2c_{nl}^2 \mathbb{E} \frac{\partial}{\partial \beta_p} \tilde{\psi}_{gl}(X_g, \beta) + 2 \sum_{l < m} c_{nm} c_{nl} \mathbb{E} \left[\tilde{\psi}_{gl}(X_g, \beta) \frac{\partial}{\partial \beta_p} \psi_{gm}(X_g, \beta) \right] \end{pmatrix} \quad (3.52)$$

$$a_n := -2 \frac{\Delta \ell}{\ell' V_n \ell} + J_n - S_n \quad (3.53)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2V_n \ell \quad (3.54)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} \Omega_n^{-1} Q_n V_n \ell. \quad (3.55)$$

Example 3, continued (*Cluster-Robust Regression*). The linear setting, with plug-in estimators, implies $B_n = 0$. We also can simplify many of the expressions making up the components of Ξ_n . Note that in this case since Q_n is invertible, $V_n = Q_n^{-1} \Omega_n Q_n^{-1}$. Furthermore, Q_n and $\hat{Q}_n$ do not depends on β or $\hat{\beta}$. Therefore, $J_n = 0$. We also have a simplification for S_n if the model is correctly specified, i.e. $\mathbb{E}[\varepsilon_g | X_g] = 0$, where X_g is the matrix with rows X'_{ig} for cluster g and let ε_g and y_g be the vector of errors and y_{ig} respectively. Then we have that $\tilde{\psi}_g(X_g, \beta) = X'_g(y_g - X'_g \beta)$. Computing S_n then gives

$$S_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \frac{\partial}{\partial \beta} (c'_n X'_g (y_g - X'_g \beta))^2 \quad (3.56)$$

$$= -\frac{2}{n} \sum_{g=1}^G \mathbb{E} c'_n X_g \varepsilon_g X'_g X_g \quad (3.57)$$

³Finding a limit distribution is still feasible in this case, but this is outside the scope of this paper; see Bentkus et al. (2007).

We then have that

$$a_n := -2 \frac{\Delta \ell}{\ell' V_n \ell} - S_n \quad (3.58)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2V_n \ell \quad (3.59)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} Q_n^{-1} \ell. \quad (3.60)$$

Constructing Y_{gn} gives

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n x_{ig} \varepsilon_{ig} \\ \sum_{i=1}^{n_g} b'_n x_{ig} x'_{ig} c_n - \sum_{i=1}^{n_g} \mathbb{E}(b_n x_{ig} x'_{ig} c_n) \\ (\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig})^2 - \mathbb{E}(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig})^2 \end{pmatrix}. \quad (3.61)$$

The result in Theorem 1 then specialises to:

$$\frac{(\ell' \hat{\beta} - \theta_0)^2}{\frac{\ell' \hat{V}_n \ell}{\sqrt{\Xi_n}} - \xi_n^2} \Rightarrow \mathcal{N}(0, 1) \quad (3.62)$$

We will compute Ξ_n now in the case of correctly specified cluster dependence, and lack of cluster dependence. First, suppose that there is no cluster dependence, i.e. $\Omega_n = \mathbb{E} x_{ig} x'_{ig} \varepsilon_{ig}^2$, however a cluster-robust variance estimator is still used. Inspection of Y_{gn} tells us that the main difference is in the third element of Y_{gn}

$$\check{Y}_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n x_{ig} \varepsilon_{ig} \\ \sum_{i=1}^{n_g} (b'_n x_{ig} \varepsilon_{ig})^2 - \sum_{i=1}^{n_g} \mathbb{E}(b'_n x_{ig} \varepsilon_{ig})^2 \\ \sum_{i=1}^{n_g} b'_n x_{ig} x'_{ig} c_n - \sum_{i=1}^{n_g} \mathbb{E}(b_n x_{ig} x'_{ig} c_n) \end{pmatrix}. \quad (3.63)$$

Under this assumption that the observations are i.i.d., i.e. no cluster-dependence, we have that under the null hypothesis, $\ell' \beta = \theta_0$,

$$\frac{n(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{Q}_n^{-1} \hat{\Omega}_n \hat{Q}_n^{-1} \ell} \Rightarrow \chi_1^2 \quad (3.64)$$

$$\frac{n(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{Q}_n^{-1} \check{\Omega}_n \hat{Q}_n^{-1} \ell} \Rightarrow \chi_1^2. \quad (3.65)$$

where $\check{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \hat{\varepsilon}_{ig}^2$ is the standard heteroskedastic-robust variance estimator. Thus, both statistics provide for valid inference under the null hypothesis, and have the same first-order asymptotic behaviour in that case. Let $\check{\Xi}_n$ be the normalising sequence when we use $\check{\Omega}_n$ in place of $\hat{\Omega}_n$. Comparing Ξ_n to $\check{\Xi}_n$ gives us an insight into differences in the asymptotic behaviour between the two procedures.

Proposition 1. *In the linear regression model, suppose a test of $H_0 : \ell' \beta = \theta_0$ can be conducted using both a cluster robust variance estimator $\hat{\Omega}_n$ and the standard heteroskedasticity-robust variance estimator $\check{\Omega}_n$:*

$$\hat{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} x_{ig} \hat{\varepsilon}_{ig} \right) \left(\sum_{i=1}^{n_g} x_{ig} \hat{\varepsilon}_{ig} \right)' \quad (3.66)$$

$$\check{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \hat{\varepsilon}_{ig}^2 \quad (3.67)$$

Then, under any fixed alternative $\theta \neq \theta_0$,

$$\frac{\frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \tilde{Q}_n^{-1} \tilde{\Omega}_n \tilde{Q}_n^{-1} \ell} - \xi_n^2}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0, 1) \quad (3.68)$$

$$\frac{\frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \tilde{Q}_n^{-1} \tilde{\Omega}_n \tilde{Q}_n^{-1} \ell} - \xi_n^2}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0, 1) \quad (3.69)$$

with $\Xi_n \geq \tilde{\Xi}_n$, with equality if and only if $\text{Var}(x_{ig} \varepsilon_{ig}) = 0$.

The difference $\Xi_n - \tilde{\Xi}_n$ can be directly compute as in the case of the t-statistic, and the difference is

$$\Xi_n - \tilde{\Xi}_n = \sum_{i \neq j} \mathbb{E}(c'_n x_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 > 0 \quad (3.70)$$

which is analogous to the t-test. If in $x_{ig} = 1$ and the ε_{ig} are i.i.d. with variance σ^2 , then this simplifies to $2(n_g - 1)\xi^2$ as in the case of the t-test.

Example 4, continued (*Quantile Regression*). In the case of quantile regression we also obtain significant simplification, this time because the Q_n matrix is estimated at a much slower rate than β and Ω_n . The reason for this is that Q_n is estimated at the $\sqrt{nh_n}$ rate, as is typical for kernel density estimators, while $\sqrt{n}(\hat{\beta} - \beta)$, $\sqrt{n}(\hat{\Omega}_n - \Omega_n) = O_P(1)$. This is also an example of when the bias term B_n is necessary to characterise the asymptotic theory. The expressions we have now are, since here Q_n is also $p \times p$, and invertible,

$$b_n = \frac{2\Delta V_n \ell}{\ell' V_n \ell} \quad (3.71)$$

$$c_n = \frac{\Delta Q_n^{-1} \ell}{\ell' V_n \ell}. \quad (3.72)$$

Here, the bias term is, recalling that our setup for this example is i.i.d.,

$$B_n = \frac{\Delta^2}{\ell' (\mathbb{E} \tilde{Q}_n \Omega^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} - \xi_n^2 \quad (3.73)$$

$$= \frac{\Delta^2}{\ell' \left(Q_n + \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) \right)^{-1} \Omega_n \left(Q_n + \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) \right)^{-1} \ell} - \xi_n^2. \quad (3.74)$$

Since the bias is tending to zero and the bias term is differentiable, asymptotically we can write that

$$B_n = b'_n \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) c_n + o(\sqrt{nh_n}) \quad (3.75)$$

$$= \frac{1}{2} b'_n \mathbb{E}(x_{ig} x'_{ig} f''(0|x_{ig}) h_n^2) c_n + o(\sqrt{nh_n}) \quad (3.76)$$

$$(3.77)$$

Since the estimator of $\hat{\beta}$ and $\hat{\Omega}$ will converge at the parametric rate, we can also simplify our expression for Ξ_n , only the variance of $\hat{Q}_n$ is relevant:

Proposition 2. Recall that R_K is the roughness of the kernel K , and $f(\cdot|x_{ig})$ and $f''(\cdot|x_{ig})$ are the conditional density of the error and the second derivative respectively. Then for the quantile regression model, we have that

$$\frac{\sqrt{nh_n}(\frac{1}{n}W_n - \xi_n^2 - B_n)}{\sqrt{\mathbb{E}[(b'_n x_{ig} x'_{ig} c_n)^2 f(0|x_{ig}) R_K]}} \Rightarrow \mathcal{N}(0, 1). \quad (3.78)$$

A direct implication of this result is that the deviations of the test statistic from the non-centrality parameter in large samples are driven by the nonparametric estimator of the variance component Q_n . This is quite different from the local asymptotic theory which treats all estimators of Q_n as equivalent, as long as they are consistent. Here, the choice of kernel and bandwidth h_n play a role in the asymptotic behaviour of the test statistic.

4 Power Calculations

We can use the central limit-theorem result in Theorem 1 to approximate power, just as we did in Section 2. Using (1) as a guide, consider the rejection probability of the standard Wald test: letting $C_{1-\alpha}^2$ be the $1 - \alpha$ -quantile of the χ_1^2 -distribution, we have that the power of a level- α test is given by:

$$P(W_n > C_{1-\alpha}^2) = P\left(\frac{\frac{1}{n}W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} > \frac{C_{1-\alpha}^2}{n\sqrt{\Xi_n}} - \frac{\xi_n^2 + B_n}{\sqrt{\Xi_n}}\right) \quad (4.1)$$

Since $\frac{1}{n}W_n - \xi_n^2 \xrightarrow{P} 0$, and the lefthand side of (4.1) is a consistent test, we must have that $(\xi_n^2 + B_n)/\sqrt{\Xi_n} \rightarrow \infty$. Thus, The only way we can approximate the power is by specifying a sequence of critical values that tends to infinity as well. We follow the case of the simple example in Section 2. Consider any sequence of the form:

$$C_n^* = n\left(\xi_n^2 + B_n - C_\beta\sqrt{\Xi_n} + \frac{C_n}{n}\right) \quad (4.2)$$

where again C_β is the β -quantile of a normal distribution. If $\Delta = \theta - \theta_0 = 0$, then note that $C_n^* = C_n$. Otherwise, rearranging the expression, we see that:

$$P(W_n > C_n^*) = P\left(\frac{\frac{1}{n}W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} > -C_\beta + \frac{C_n}{n\sqrt{\Xi_n}}\right) \quad (4.3)$$

Thus, we have that $P(W_n > C_n^*) \rightarrow 1 - \beta$. C_n/n is not relevant for the present theory, but when we consider computing power over a range of values of ξ_n^2 , it may be useful include to nest the case that if $\xi_n^2 = 0$ we can set $C_n = C_{1-\alpha}^2$ and still compute the approximate power function. Notice that if we use (4.3) for sample-size calculations, as we do in Section 5, Ξ_n plays a role in reducing the power, or rather increasing the sample size required to achieve a particular power. Thus, results such as Proposition 1 can be translated into results on power. We set C_n in the numerical exercises so that C_n^* leads to size control, and we will see that the size of Ξ_n has an impact on what sample size is necessary to achieve a particular power.

We now return to our discussion of Bahadur asymptotic relative efficiency and asymptotic power calculations. To make such a connection, we must make a connection with Type-I error and power in our present setup. We can in fact provide a slight generalisation to Proposition 2 in Bahadur (1960b):

Proposition 3. Suppose that for a sequence of tests the power $P(W_n > C_n)$ converges to a constant

$1 - \beta \in (0, 1)$. Let $\alpha_n := 1 - F_{0,n}(C_n)$ denote the Type-I error for the test. Then, it must be that

$$\frac{\log \alpha_n}{\log(1 - F_{0,n}(W_n))} \rightarrow 1. \quad (4.4)$$

Furthermore, suppose we have two tests, $W_n^{(1)}$ and $W_n^{(2)}$ with CDFs under the null hypothesis $F_{n,0}^{(1)}$ and $F_{n,0}^{(2)}$. Then, for any two sequences $C_n^{(1)}, C_n^{(2)}$ such that $P(W_n^{(i)} > C_n^{(i)}) \rightarrow 1 - \beta$ for each i , we have that the Bahadur asymptotic relative efficiency is given by

$$\lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^{(1)}(C_n^{(1)}))}{\log(1 - F_{n,0}^{(2)}(C_n^{(2)}))} \quad (4.5)$$

This result is also similar to Lemma 1 and the subsequent Corollary in the Appendix of Groeneboom and Oosterhoff (1981). The main difference between the statement of Proposition 3 and Proposition 2 in Bahadur (1960b) is that there is no assumption here that the correct normalising sequence for $\log \alpha_n$ is $\frac{1}{n}$, however modifying that proof to accommodate this is trivial. The proposition is straightforward, but we did not find its statement in the literature. The second part follows from the first immediately as the Bahadur asymptotic relative efficiency is defined as

$$\lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^{(1)}(W_n^{(1)}))}{\log(1 - F_{n,0}^{(2)}(W_n^{(2)}))} \quad (4.6)$$

if it exists. Rearranging and normalising by $\log \alpha_n^{(i)} = \log(1 - F_{n,0}^{(i)}(C_n^{(i)}))$, then using (4.4) leads to the result. Proposition 3 gives a recipe for finding the right sequence of tests so that the p-values under a particular alternative and Type-I error go to zero at the same rate.

1. For a test statistic W_n that can be used to test $H_0 : \mu = \mu_0$, choose an alternative $\mu \neq \mu_0$.
2. Find a sequence of critical values C_n such that under this alternative $P(W_n > C_n) \rightarrow 1 - \beta \in (0, 1)$.
3. The Type-I error rate using this sequence of critical values converges to 0 at the same rate as the p-values under the alternative $\mu \neq \mu_0$.

One difference to point out between the Bahadur approach and our power calculations is that the choice of $1 - \beta$ is irrelevant for finding the rate at which $\alpha_n \rightarrow 0$: the p-values do not depend on the critical values, and therefore only the particular alternative is relevant for the Bahadur comparison. In general, Bahadur asymptotic relative efficiency calculations are done with the choice $C_n = n\xi_n^2$. In our setup, this is equivalent to specifying a sequence of tests such that power converges to 1/2 in most applications. Our results imply the following interpretation of Bahadur asymptotic relative efficiency: a test is preferred in the Bahadur sense if it can achieve smaller size than another test when both tests have power 1/2. One notable exception is when $\frac{1}{n}W_n$ is consistent for ξ_n^2 but possibly asymptotically biased, which is covered in our case with B_n .

For the simple example using the t-test, we can directly compute the rate at which size tends to 0 for the choices of critical values. Proposition 3 implies that we can set $C_\beta = 0$ to find the rate at which size tends

to 0. For this part, we assume that the X_{ig} are i.i.d. $\mathcal{N}(\mu, 1)$. Doing so for W_n^h and W_n^c leads to

$$-\frac{1}{n} \log(1 - F_{n,0}^h(n\xi^2)) \rightarrow \frac{1}{2} \log(1 + \xi^2) \quad (4.7)$$

$$-\frac{1}{n} \log(1 - F_{n,0}^c(n\xi^2)) \rightarrow \frac{1}{2n_g} \log\left(1 + \frac{\xi^2}{n_g}\right) \quad (4.8)$$

This comparison gives the convergence rate of Type-I error to zero under a sequence of tests which lead to power converging to 1/2. If $n_g = 1$ the two expressions are the same. The ratio

$$\frac{n_g \log(1 + \xi^2)}{\log(1 + \xi^2/n_g)} \quad (4.9)$$

is maximised at $\xi^2 = 0$, tends to n_g as $\xi^2 \rightarrow \infty$, and it can be shown that this ratio monotonically decreasing in ξ^2 and thus is always greater than 1, uniformly in ξ^2 , as long as $n_g > 1$.

A simple way of identifying the preference for the homoskedastic test using W_n^h can be seen based on the asymptotic power analysis. Based on the above calculations for the Type-I error, we have that

$$\frac{\log(1 - F_{n,0}^h(n\xi^2))}{\log(1 - F_{n,0}^c(nn_g((\xi^2 + 1)^{n_g} - 1)))} \rightarrow 1 \quad (4.10)$$

To see this, observe that

$$\frac{1}{2n_g} \log\left(1 + \frac{n_g((\xi^2 + 1)^{n_g} - 1)}{n_g}\right) = \frac{1}{2} \log(1 + \xi^2)$$

Thus, if we specify the tests as rejecting when $W_n^h > n\xi^2$ and $W_n^c > nn_g((\xi^2 + 1)^{n_g} - 1)$, we can compare the tests with respect to power, as Type-I error converges to zero for each test at the same rate. We have that

$$P(W_n^h > n\xi^2) \rightarrow 1/2 \quad (4.11)$$

and thus need only consider W_n^c . Computing the power function using our approximation, we have

$$\begin{aligned} P(W_n^c > nn_g((\xi^2 + 1)^{n_g} - 1)) &= P\left(\sqrt{n}\left(\frac{1}{n}W_n^c - \xi^2\right) > \sqrt{n}(n_g(\xi^2 + 1)^{n_g} - (\xi^2 + 1))\right) \\ &\rightarrow 0. \end{aligned}$$

We can generalise this in the following proposition:

Proposition 4. *Under the assumption that the data X_{ig} are i.i.d. $\mathcal{N}(\mu, 1)$, with $\xi^2 = (\mu - \mu_0)^2$, for any sequences of critical values C_n^h, C_n^c such that $P(W_n^h > C_n^h) \rightarrow 1 - \beta \in (0, 1)$ and*

$$\frac{\log(1 - F_{n,0}^h(C_n^h))}{\log(1 - F_{n,0}^c(C_n^c))} \rightarrow 1 \quad (4.12)$$

we have that

$$P(W_n^c > C_n^c) \rightarrow 0. \quad (4.13)$$

To see that the proposition is true, consider first that determining the convergence rate of Type-I error to zero does not depend on the asymptotic power $1 - \beta$, by Proposition 3. Thus, for any C_n^h , we can use the

choice $C_n^c = nn_g((\xi^2 + 1)^{n_g} - 1)$. For this sequence, $P(W_n^c > C_n^c) \rightarrow 0$, and thus the proposition is proved.

This analysis might suggest that considering the power of the tests as size tends to zero does not necessarily add new insights in the comparison of tests beyond that of the Bahadur approach. In general, the standard goal of relative efficiency is different from power analysis: try to understand the necessary sample size to attain a particular size and power combination for a given alternative. Bahadur asymptotic relative efficiency approximates the necessary sample size to achieve power $1 - \beta$ at alternative $\mu \neq \mu_0$ by sending size to 0. Let $n_a(\alpha, \beta, \mu)$ be the sample size necessary for $P(W_n^a > C_n^a) \geq 1 - \beta$ with $1 - F_{0,n}^a(C_n^a) \leq \alpha_n$. Along the chosen sequence, $\alpha_n \rightarrow 0$. Then, Bahadur approximates the relative efficiency $n_c(\alpha_n, \beta, \mu)/n_h(\alpha_n, \beta, \mu)$ based on the fact that

$$\lim_{n \rightarrow \infty} \frac{n_c(\alpha_n, \beta, \mu)}{n_h(\alpha_n, \beta, \mu)} = \lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^h(C_n^h))}{\log(1 - F_{n,0}^c(C_n^c))} = \lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^h(W_n^h))}{\log(1 - F_{n,0}^c(W_n^c))}.$$

In all of these expressions, β drops out, but as discussed in Groeneboom and Oosterhoff (1981), in general computing $n_c(\alpha_n, \beta, \mu)/n_h(\alpha_n, \beta, \mu)$ will depend on β . Thus, one can understand the computation of the limiting distribution of the test statistics under a fixed alternative as providing a clarification of how to interpret Bahadur asymptotic relative efficiency. We can also consider the analysis in Kallenberg (1981), in which they considered the Bahadur deficiency, $n_c(\alpha_n, \beta, \mu) - n_h(\alpha_n, \beta, \mu)$. In that paper, Theorem 1.2 gives an expression for the power of the test in terms of the p-values. They point out that the deficiency still depends on β in general. We show in the simulation section that sample size calculations using our setup can provide more accurate results, as they allow for the resulting sample size to depend on β .

5 Numerical results

In this section we present a set of numerical results for the simple example from Section 2 in which we compute the sample size required for a particular power. For each data-generating process we fix size and compute the minimum sample size required to achieve a particular power. For Pitman, with $W_n \Rightarrow \chi_1^2(\xi^2)$, we will use the $\chi_1^2(n\xi^2)$ distribution to come up with the estimated sample size. For our approximation, we will use n^* as a solution to

$$n^* = \left(\Phi^{-1}(\beta) + \frac{\xi_{n^*}^2 + B_{n^*}}{\sqrt{\Xi_{n^*}}} \right) \frac{\sqrt{\Xi_{n^*}}}{C_n} \quad (5.1)$$

where in each case we will choose C_n such that the test has size α under the null. Note that we cannot use Bahadur to perform this exercise.

5.1 t-test

We first present some numerical results on the t-test. All results in these tables were computed numerically, without relying on simulation. We consider the simple t-test and compare a statistics formed using cluster-robust variance estimator to a homoskedastic variance estimator. We vary the size, power, alternative, and cluster size across the various calculations. In this setup, we know the distribution of the Wald-statistics.

Recall our two test statistics from Section 2:

$$W_h = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_h^2}, \quad \hat{\sigma}_h^2 = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} (X_{gi} - \bar{X}_n)^2$$

$$W_c = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_c^2}, \quad \hat{\sigma}_c^2 = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{gi} - \bar{X}_n) \right)^2.$$

The homoskedastic statistic W_h (re-scaled) is $F_{1,n-1}$ distributed under the null, and $F_{1,n-1}(n\xi^2)$ under the alternative. The cluster-robust test statistic W_c (re-scaled) is $F_{1,G-1}$ distributed under the null hypothesis and $F_{1,G-1}(n\xi^2)$ under the alternative. The observations are i.i.d. $\mathcal{N}(\theta, 1)$, where $\theta = 0$ under the null and $\theta \in \{0.5, 0.8, 1.0\}$ under the alternative. We test the hypothesis $H_0 : \theta = 0$ at the 0.05 level and the 0.01 level. We aim to achieve power $1 - \beta \in \{0.8, 0.9\}$.

The results are summarised in Tables 1-4.

θ	n_g	N_h	N_c	$\hat{N}_p$	$\hat{N}_{*,h}$	$\hat{N}_{*,c}$	$ERR_{p,h}$	$ERR_{p,c}$	$ERR_{*,h}$	$ERR_{*,c}$
0.50	10.00	33.37	53.51	33.14	38.58	92.62	0.01	0.38	-0.16	-0.73
0.80	10.00	14.30	35.04	14.07	16.99	37.51	0.02	0.60	-0.19	-0.07
1.00	10.00	9.94	30.52	9.71	11.98	24.79	0.02	0.68	-0.21	0.19
0.50	20.00	33.37	76.83	33.14	38.58	150.03	0.01	0.57	-0.16	-0.95
0.80	20.00	14.30	57.27	14.07	16.99	59.76	0.02	0.75	-0.19	-0.04
1.00	20.00	9.94	51.95	9.71	11.98	38.93	0.02	0.81	-0.21	0.25

Table 1: $\alpha = 0.05$, $1 - \beta = 0.8$.

θ	n_g	N_h	N_c	$\hat{N}_p$	$\hat{N}_{*,h}$	$\hat{N}_{*,c}$	$ERR_{p,h}$	$ERR_{p,c}$	$ERR_{*,h}$	$ERR_{*,c}$
0.50	10.00	50.06	80.87	49.63	55.00	111.96	0.01	0.39	-0.10	-0.38
0.80	10.00	21.63	51.48	21.21	24.18	45.58	0.02	0.59	-0.12	0.11
1.00	10.00	15.08	44.03	14.67	17.03	30.27	0.03	0.67	-0.13	0.31
0.50	20.00	50.06	113.93	49.63	55.00	170.39	0.01	0.56	-0.10	-0.50
0.80	20.00	21.63	81.77	21.21	24.18	68.09	0.02	0.74	-0.12	0.17
1.00	20.00	15.08	72.84	14.67	17.03	44.50	0.03	0.80	-0.13	0.39

Table 2: $\alpha = 0.01$, $1 - \beta = 0.8$.

θ	n_g	N_h	N_c	$\hat{N}_p$	$\hat{N}_{*,h}$	$\hat{N}_{*,c}$	$ERR_{p,h}$	$ERR_{p,c}$	$ERR_{*,h}$	$ERR_{*,c}$
0.50	10.00	44.00	64.20	43.53	57.16	177.60	0.01	0.32	-0.30	-1.77
0.80	10.00	18.45	39.71	17.97	25.09	71.63	0.03	0.55	-0.36	-0.80
1.00	10.00	12.59	33.89	12.11	17.66	47.17	0.04	0.64	-0.40	-0.39
0.50	20.00	44.00	88.23	43.53	57.16	309.40	0.01	0.51	-0.30	-2.51
0.80	20.00	18.45	63.00	17.97	25.09	123.01	0.03	0.71	-0.36	-0.95
1.00	20.00	12.59	56.38	12.11	17.66	79.99	0.04	0.79	-0.40	-0.42

Table 3: $\alpha = 0.05$, $1 - \beta = 0.9$.

In each table, θ is the alternative, n_g is the cluster size, N_h is the sample size required for W_h to have power $1 - \beta$ at level α , and similarly N_c is the sample size required for W_c to have the same power. $\hat{N}_p$, $\hat{N}_{*,h}$ and $\hat{N}_{*,c}$ are the estimated sample sizes using local asymptotic power in the style of Pitman (1949), the proposed method in this paper for W_h , and the proposed method for W_c , respectively. $ERR_{p,h}$ and

θ	n_g	N_h	N_c	$\hat{N}_p$	$\hat{N}_{*,h}$	$\hat{N}_{*,c}$	$ERR_{p,h}$	$ERR_{p,c}$	$ERR_{*,h}$	$ERR_{*,c}$
0.50	10.00	62.87	94.26	62.10	75.07	198.24	0.01	0.34	-0.19	-1.10
0.80	10.00	26.64	57.57	25.88	32.81	80.03	0.03	0.55	-0.23	-0.39
1.00	10.00	18.30	48.47	17.55	23.00	52.75	0.04	0.64	-0.26	-0.09
0.50	20.00	62.87	128.67	62.10	75.07	330.72	0.01	0.52	-0.19	-1.57
0.80	20.00	26.64	89.37	25.88	32.81	131.55	0.03	0.71	-0.23	-0.47
1.00	20.00	18.30	78.74	17.55	23.00	85.59	0.04	0.78	-0.26	-0.09

Table 4: $\alpha = 0.01, 1 - \beta = 0.9$.

$ERR_{p,c}$ are the relative error in the local power approximation: $(N_a - \hat{N}_p)/N_a$. Note that for the local power approximation, there is no distinction between the two test statistics. Similarly, $ERR_{*,h}$ and $ERR_{*,c}$ are the relative errors when using the proposed method in this paper.

Not surprisingly, local asymptotic power is quite accurate for estimating N_h ; power calculations using the noncentral χ^2_1 distribution and the noncentral $F_{1,n-1}$ will be very similar for any moderate sample size. What is interesting is that our approximation, while not achieving the same accuracy as local power, achieves similar error rates to the local asymptotic approximation to the cluster-robust statistic, even when size is relative large, as in Table 1.

Looking at Table 1 in more detail, our approximation does much better than local power in performing the power calculations for W_c , except in the case when $\theta = 0.50$. We would expect that the local power approximation dominates the procedure presented in this paper for smaller values of the alternative, and this is what we see here. Comparing Table 1 to 2, overall the fixed-alternative approximation improves as α decreases from 0.05 to 0.01, and is uniformly more accurate than the local power approximation of W_c . What is interesting is that the cluster-sizes, which drive the difference in the asymptotic distribution of the two statistics, do not seem to drastically affect the accuracy of the local power calculations. Local power does not vary much in accuracy across the tables. The normal approximation in this paper does vary quite a bit in terms of accuracy. The sample size calculation is much more accurate when θ is relatively large, and is much less accurate for small θ than the local power calculations.

6 Conclusion

In this paper we develop a first-order asymptotic theory of Wald test statistics under fixed alternatives. We motivate this discussion by mapping the asymptotic distribution to a relative efficiency measure. Our main finding is that this alternative asymptotic framework distinguishes between approaches to testing that more classical approaches cannot. This opens up the possibility of comparing different variance estimators with testing in mind. This contrasts with previous comparisons that have previously been made based on simulation evidence, higher-order comparisons, or finite-sample criteria. Our approach applies to a broad class of models. One conclusion of particular interest for applied researchers is that there is an asymptotic justification for assuming there is a cost for clustering at too-coarse a level.

There are also plenty of cases of interest not considered here. Our analysis could be applied to comparing commonly used heteroskedastic-robust variance estimators. We also did not pursue any high-dimensional or machine learning applications here, and it would be interesting to consider how our power analysis could provide guidance for tuning parameter choices in that setting. We could also extend the results to accomodate settings where the variance estimator is not asymptotically normal, as is the case when the 4th moment of the influence function does not exist. The theory in Bentkus et al. (2007) introduces similar results in the

case of the 1-sample t-test, and similar results could be produced in a more general setting, such as in the case of linear processes driven by i.i.d. infinite-variance innovations (see, for example, Cavaliere, Georgiev, and Taylor (2016)).

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A Proofs of Main Results

A.1 Proof of Theorem 1

Proof. The delta-method can be used to find the asymptotic distribution of

$$\frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{V}_n \ell}. \quad (\text{A.1})$$

Recall $\tilde{Q}_n$ and define $\tilde{\Omega}$:

$$\tilde{Q}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \beta) \quad (\text{A.2})$$

$$\tilde{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)'. \quad (\text{A.3})$$

We first expand to isolate the variance estimator:

$$\begin{aligned} \frac{(\ell' \hat{\beta} - \theta_0)^2}{\ell' \hat{V}_n \ell} - \xi_n^2 - B_n &= \frac{(\ell' \hat{\beta} - \theta)^2}{\ell' \hat{V}_n \ell} - 2 \frac{\Delta(\ell' \hat{\beta} - \theta)}{\ell' \hat{V}_n \ell} + \frac{\Delta^2}{\ell' \hat{V}_n \ell} - \xi_n^2 - B_n \\ &= \frac{(\ell' \hat{\beta} - \theta)^2}{\ell' \hat{V}_n \ell} - 2 \frac{\Delta(\ell' \hat{\beta} - \theta)}{\ell' \hat{V}_n \ell} \\ &\quad + \frac{\Delta^2}{\ell' (\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell' (Q'_n \Omega^{-1} Q_n)^{-1} \ell} - B_n \\ &\quad + \frac{\Delta^2}{\ell' (\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell' (\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \\ L_{1n} &= \frac{(\ell' \hat{\beta} - \theta)^2}{\ell' \hat{V}_n \ell} - 2 \frac{\Delta(\ell' \hat{\beta} - \theta)}{\ell' \hat{V}_n \ell} \\ L_{2n} &= \frac{\Delta^2}{\ell' (\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell' (Q'_n \Omega^{-1} Q_n)^{-1} \ell} - B_n \\ L_{3n} &= \frac{\Delta^2}{\ell' (\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell' (\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell}. \end{aligned}$$

L_{1n} is the contribution of $\hat{\beta}$, L_{2n} is the contribution of $\hat{Q}_n$, and L_{3n} is the contribution of $\hat{\Omega}_n$.

For L_{1n} , we have that

$$\sqrt{\ell' V_n \ell} \sqrt{n} L_{2n} = - \frac{2\Delta}{\sqrt{\ell' V_n \ell}} \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \ell' (Q'_n \Omega_n^{-1} Q_n)^{-1} Q'_n \Omega_n^{-1} \psi(X_{ig}, \beta) + o_P(1) \quad (\text{A.4})$$

by (3.15). Define

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2V_n \ell \quad (\text{A.5})$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} \Omega_n^{-1} Q_n V_n \ell. \quad (\text{A.6})$$

Now, focusing on L_{2n} , applying the delta-method gives us

$$\frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(Q'_n \Omega^{-1} Q_n)^{-1} \ell} - B_n = \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \quad (\text{A.7})$$

$$- \frac{\Delta^2}{\ell'(\mathbb{E} \tilde{Q}'_n \Omega^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} \quad (\text{A.8})$$

$$= \frac{\Delta^2}{(\ell' V_n \ell)^2} \text{tr}(2V_n \ell \ell' V_n Q'_n \Omega_n^{-1} (\hat{Q}_n - \mathbb{E} \tilde{Q}_n)) \quad (\text{A.9})$$

$$+ o_P(\|\hat{Q}_n - \mathbb{E} \tilde{Q}_n\|) \quad (\text{A.10})$$

$$= b'_n(\hat{Q}_n - \mathbb{E} \tilde{Q}_n)c_n + o_P(|b'_n(\hat{Q}_n - \mathbb{E} \tilde{Q}_n)c_n|) \quad (\text{A.11})$$

Note that we can rewrite

$$\hat{Q}_n - \mathbb{E} \tilde{Q}_n = \tilde{Q}_n - \mathbb{E} \tilde{Q}_n \quad (\text{A.12})$$

$$+ \mathbb{E} \hat{Q}_n - \mathbb{E} \tilde{Q}_n \quad (\text{A.13})$$

$$+ [\hat{Q}_n - \mathbb{E} \hat{Q}_n - (\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)] \quad (\text{A.14})$$

By (3.25),

$$\|[\hat{Q}_n - \mathbb{E} \hat{Q}_n - (\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)]\| = o_P(\|\hat{\beta} - \beta\| + \|\tilde{Q}_n - \mathbb{E} \tilde{Q}_n\|) \quad (\text{A.15})$$

Therefore, if we define

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c'_n \frac{\partial}{\partial \beta_1} \mathbb{E} \hat{\psi}(X_{ig}, \beta) b_n \\ \vdots \\ c'_n \frac{\partial}{\partial \beta_p} \mathbb{E} \hat{\psi}(X_{ig}, \beta) b_n \end{pmatrix} \quad (\text{A.16})$$

we can write L_{2n} as

$$L_{2n} = b'_n(\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)c_n + J'_n(\hat{\beta} - \beta) + o_P(|J'_n(\hat{\beta} - \beta)| + |b'_n(\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)c_n|) \quad (\text{A.17})$$

Lastly, working with L_{3n} , we have

$$L_{3n} = \frac{\Delta^2}{\ell'(\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \quad (\text{A.18})$$

$$= -\frac{\Delta^2}{(\ell' V_n \ell)^2} \text{tr}(\Omega_n^{-1} Q_n V_n \ell \ell' V_n Q'_n \Omega_n^{-1} (\hat{\Omega}_n - \Omega_n)) + o_P(\|\hat{\Omega}_n - \Omega_n\|) \quad (\text{A.19})$$

$$= -c'_n(\hat{\Omega}_n - \Omega_n)c_n + o_P(|c'_n(\hat{\Omega}_n - \Omega_n)c_n|) \quad (\text{A.20})$$

Defining

$$\tilde{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' \quad (\text{A.21})$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} c'_n \frac{\partial}{\partial \beta_1} \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' c_n \\ \vdots \\ c'_n \frac{\partial}{\partial \beta_p} \mathbb{E} \tilde{\psi}_g(X_g, \beta) \tilde{\psi}_g(X_g, \beta)' c_n \end{pmatrix}. \quad (\text{A.22})$$

we then write

$$L_{3n} = -c'_n(\tilde{\Omega}_n - \Omega_n)c_n - S'_n(\hat{\beta} - \beta) + o_P(|S'_n(\hat{\beta} - \beta)|) + o_P(|c'_n(\hat{\Omega}_n - \Omega_n)c_n|) \quad (\text{A.23})$$

Therefore, defining $a_n = -2\frac{\Delta\ell}{\ell'\hat{V}_n\ell} + J_n - S_n$, we have shown that

$$\frac{(\ell'\hat{\beta} - \theta_0)^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n = a'_n(\hat{\beta} - \beta) + \quad (\text{A.24})$$

$$+ c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n \quad (\text{A.25})$$

$$- c'_n(\tilde{\Omega}_n - \Omega_n)c_n \quad (\text{A.26})$$

$$+ o_P(|a'_n\hat{\beta} - \beta|) + o_P(|b'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)c_n|) + o_P(|c'_n\hat{\Omega}_n - \Omega_n)c_n|. \quad (\text{A.27})$$

By Assumption 3, we can ignore these o_P terms if we find a normalising sequence for the other terms: that is to say if we find a sequence Ξ_n such that

$$\Xi_n^{-1/2} \left(a'_n(\hat{\beta} - \beta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n \right) = O_P(1) \quad (\text{A.28})$$

We have that

$$\Xi_n^{-1/2} \left(\frac{(\ell'\hat{\beta} - \theta_0)^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n - \left(a'_n(\hat{\beta} - \beta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n \right) \right) = o_P(1) \quad (\text{A.29})$$

We will use Theorem 2 in Hansen and Lee (2019). Define

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n(Q'_n\Omega_n^{-1}Q_n)^{-1}Q'_n\Omega_n^{-1}\psi(X_{ig}, \beta) \\ \sum_{i=1}^{n_g} c'_n(\hat{\psi}(X_{ig}, \beta) - \mathbb{E}\hat{\psi}(X_{ig}, \beta))b_n \\ -c'_n(\tilde{\psi}_g(X_g, \beta)\tilde{\psi}_g(X_g, \beta)' - \mathbb{E}\tilde{\psi}_g(X_g, \beta)\tilde{\psi}_g(X_g, \beta)')c_n \end{pmatrix} \quad (\text{A.30})$$

Note that $\mathbb{E}Y_{gn} = 0$, $\frac{1}{n}\sum_{g=1}^G Y_{gn} = a'_n(\hat{\beta} - \beta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n$, and define $\Xi_n = \mathbb{E}\sum_{g=1}^G (\mathbf{1}'Y_{gn})^2$. The uniform integrability condition in Assumption 2 and Assumption 1 gives us that

$$\Xi_n^{-1/2} \sum_{g=1}^G \mathbf{1}'Y_{gn} \Rightarrow \mathcal{N}(0, 1) \quad (\text{A.31})$$

by Theorem 2 in Hansen and Lee (2019), with the assumption that $\mathbf{1}'Y_{gn}$ is nondegenerate for all n, g . Finally, by Assumption (3),

$$\Xi_n^{-1/2} \left(\frac{1}{n}W_n - \xi^2 - B_n - \sum_{g=1}^G Y_{gn} \right) = o_P(1). \quad (\text{A.32})$$

□

A.2 Proof of Proposition 1

Proof. Let $\tilde{\Xi}_n = \mathbb{E}(\mathbf{1}'\tilde{Y}_{ng})^2$, where $\tilde{Y}_{ng}$ is formed using $\tilde{\Omega}_n$ in place of $\hat{\Omega}_n$. When computing $\Xi_n - \tilde{\Xi}_n$, we obtain a great simplification that when looking at $\mathbb{E}Y_{gn}Y'_{gn} - \mathbb{E}\tilde{Y}_{gn}\tilde{Y}'_{gn}$, we note that all terms are zero,

except for the second diagonal element. We can compute this component as a function of the moments of

$$[\mathbb{E} Y_{gn} Y'_{gn}]_{2,2} - [\mathbb{E} \check{Y}_{gn} \check{Y}'_{gn}]_{2,2} = \sum_{i \neq j} \mathbb{E}(c'_n x'_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 \geq 0 \quad (\text{A.1})$$

where we use the notation $[\cdot]_{a,b}$ to denote the (a,b) element of the relevant matrix. To see why (A.1) is true, notice that the components of $[\check{Y}_{gn}]_2$ and $[Y_{gn}]_2$ are mean-zero. However, the variance for the second component of $\check{Y}_{gn}$ is

$$\text{Var} \left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig})^2 \right) = n_g \text{Var}((c'_n x_{ig} \varepsilon_{ig})^2). \quad (\text{A.2})$$

Under Assumption 1, the cluster-robust variance estimator is still consistent. However, now we must assess the variances of these variance estimators:

$$\text{Var} \left(\left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^2 \right) = \mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 - \left(\mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^2 \right)^2 \quad (\text{A.3})$$

$$= \mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 - n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2. \quad (\text{A.4})$$

We can write this fourth-moment in terms of the fourth cumulant, which we will denote $k_4(\cdot)$, and use the property that the cumulant of the sum is the sum of cumulants:

$$\mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 = k_4 \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.5})$$

$$= \sum_{i=1}^{n_g} k_4(c'_n x_{ig} \varepsilon_{ig}) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.6})$$

$$= n_g k_4(c'_n x_{ig} \varepsilon_{ig}) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.7})$$

Thus, note that:

$$[\mathbb{E} Y_{gn} Y'_{gn}]_{2,2} - [\mathbb{E} \check{Y}_{gn} \check{Y}'_{gn}]_{2,2} = \text{Var} \left(\left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig}) \right)^2 \right) - \text{Var} \left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig})^2 \right) \quad (\text{A.8})$$

$$= n_g k_4(c'_n x_{ig} \varepsilon_{ig}) + 2n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.9})$$

$$- n_g k_4(c'_n x_{ig} \varepsilon_{ig}) - 2n_g \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.10})$$

$$= 2n_g(n_g - 1) \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.11})$$

$$= \sum_{i \neq j} \mathbb{E}(c'_n x_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 > 0 \quad (\text{A.12})$$

This expression is completely analogous to the case with the sample mean. $\square$

A.3 Proof of Proposition 2

Proof. As a result of Lemma 2, and since we know that $\hat{\beta} - \beta = O_P(1/\sqrt{n})$ and $\hat{\Omega}_n - \Omega_n = O_P(1/\sqrt{n})$. Standard results on kernel density estimation lead to the expression for B_n and that for any matrix A we have that:

$$\sqrt{nh_n} \left(\frac{1}{n} \sum_{i=1}^n b'_n x_{ig} x'_{ig} c_n \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) - \xi^2 - B_n \right) \Rightarrow \mathcal{N} \left(0, \mathbb{E}(b'_n x_{ig} x'_{ig} c_n)^2 f(0|x_{ig}) R_K \right) \quad (\text{A.1})$$

□

B Proofs of Lemmas

B.1 Proof of Lemma 1

A Taylor expansion gives us that, for $\hat{\beta} \in N_\beta$.

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{\partial}{\partial \beta} \psi(X_{ig}, \hat{\beta}) - \dot{\psi}(\hat{\beta}) - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{\partial}{\partial \beta} \psi(X_{ig}, \beta) - \dot{\psi}(\beta) \quad (\text{B.1})$$

$$= \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \sum_{k=1}^p \left(\frac{\partial^2}{\partial \beta \partial \beta'} \psi_k(X_{ig}, \bar{\beta}) - \mathbb{E} \frac{\partial^2}{\partial \beta \partial \beta'} \psi_k(X_{ig}, \bar{\beta}) \right)' (\hat{\beta} - \beta) \quad (\text{B.2})$$

$$= o_P(1) O_P(1/\sqrt{n}) = o_P(1/\sqrt{n}) \quad (\text{B.3})$$

by the assumption of bounded derivatives and the fact that $\sqrt{n}(\hat{\beta} - \beta) = O_P(1)$. Since $P(\hat{\beta} \in N_\beta) \rightarrow 1$, for any neighborhood N_β , the result follows. Similarly, for

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\beta}) \psi(X_{ig}, \hat{\beta})' - \Psi(\hat{\beta}) - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \beta) \psi(X_{ig}, \beta)' - \Psi(\beta) \quad (\text{B.4})$$

$$= \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \bar{\beta})' \frac{\partial}{\partial \beta} \psi(X_{ig}, \bar{\beta})' - \mathbb{E} \left(\psi(X_{ig}, \bar{\beta})' \frac{\partial}{\partial \beta} \psi(X_{ig}, \bar{\beta})' \right) \right) (\hat{\beta} - \beta) \quad (\text{B.5})$$

$$= o_P(1/\sqrt{n}) \quad (\text{B.6})$$

B.2 Proof of Lemma 2

Proof. Boundedness of the kernel function K and the existence of integrable functions G_j which uniformly bound the error density and the derivatives implies that

$$\mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) = O_P(1/\sqrt{n}) \quad (\text{B.1})$$

via the delta-method. Now, we focus on:

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) \quad (\text{B.2})$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \right) \quad (\text{B.3})$$

$$= \frac{1}{\sqrt{n}} (\mathbb{G}_n(x x' \hat{f}(\hat{\beta})) - \mathbb{G}_n(x x' \hat{f}(\beta))) \quad (\text{B.4})$$

using the common notational shorthand $\mathbb{G}_n \psi(\beta) = \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \beta) - \mathbb{E} \psi(X_{ig}, \beta)$. An implication of Assumption 3^{reg} is that there exist functions K_1, K_2 such that K_{ig} is non-negative, non-decreasing, and $K = K_1 - K_2$. Furthermore, $|K|_v = |K_1|_v + |K_2|_v$, so we have a simple form of the total-variation norm. Using arguments similar to those found in Einmahl and Mason (2000), we have that, for $t, s \in \mathbb{R}^p$, letting $\delta_t = t - \beta, \delta_s = s - \beta$,

$$\begin{aligned} K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) &= K_1 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K_1 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \\ &\quad - \left(K_2 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K_2 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right) \\ &= \int_{\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}}^{\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n}} dK_1(x) - \int_{\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}}^{\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n}} dK_2(x) \end{aligned}$$

This implies, via the triangle inequality,

$$\left| K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right| \leq \int \left| \mathbb{1}_{\left[\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}, \frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right]}(x) \right| d(K_1(x) + K_2(x)) \quad (\text{B.5})$$

Thus, using (B.5), we can use Hölder's inequality to bound the mean-squared difference:

$$\mathbb{E} \left[\left(K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right)^2 \middle| x_{ig} \right] \quad (\text{B.6})$$

$$\leq \int \mathbb{E} \left| \mathbb{1}_{\left[\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}, \frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right]}(x) \right| d(K_1(x) + K_2(x)) |K|_v \quad (\text{B.7})$$

$$= \int \left| \int_{x'_{ig} \delta_s + h_n x}^{x'_{ig} \delta_t + h_n x} f(\varepsilon | x_{ig}) d\varepsilon \right| d(K_1(x) + K_2(x)) |K|_v \quad (\text{B.8})$$

$$\leq \|f(\cdot | x_{ig})\|_\infty |K|_v^2 \|x_{ig}\|_2 \|t - s\|_2 \quad (\text{B.9})$$

Now, by Assumption 3^{reg}:

$$\mathbb{E} \left[\left(K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right)^2 \right] = O(\|t - s\|_2) \quad (\text{B.10})$$

Putting this all together, for any $\delta > 0$, let $N_{\delta/\sqrt{n}}(\beta)$ be a $\delta/\sqrt{n}$ neighborhood of β . Then, we have that

for any $\epsilon > 0$,

$$P \left(\sup_{b \in N_{\delta/\sqrt{n}}(\beta)} \left| \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\beta))) \right| > h_n^{1/2}\epsilon \right) \quad (\text{B.11})$$

$$\leq \frac{1}{\epsilon h_n^{1/2}} \mathbb{E} \left(\sup_{b \in N_{\delta/\sqrt{n}}(\beta)} \left| \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\beta))) \right| \right) \quad (\text{B.12})$$

We now need a slight extension of a VC-class result from Giné and Nickl (2016):

Lemma 3. *Let $\mathcal{K} = \{(\varepsilon, x) \mapsto K\left(\frac{\varepsilon - x't}{h}\right) : t \in \mathbb{R}^p, h > 0\}$. Then $\mathcal{K}$ is of VC-type.*

The arguments are the same as in Giné and Nickl (2016), with the finite-dimensional vector space having dimension $p + 2$, so we omit the proof.

We are now ready to use the maximal inequality of Chernozhukov, Chetverikov, and Kato (2014):

$$h_n^{-1/2} \mathbb{E} \left(\sup_{b \in N_{\delta/\sqrt{n}}(\beta)} \left| \mathbb{G}_n(\psi_b - \psi_\beta) \right| \right) = O \left(\sqrt{\frac{\log n}{h_n n^{1/2}}} \right)$$

where in the notation of Corollary 5.1 of Chernozhukov et al. (2014), we can choose $\sigma^2 = O(1/\sqrt{n})$ by (B.10). This means that when $h_n = o(\log n/\sqrt{n})$, we obtain the desired result:

$$\frac{1}{\sqrt{n}} \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\beta))) = o_P(1/\sqrt{nh_n}) \quad (\text{B.13})$$

□